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Causal inference and explainable artificial intelligence based quantum deep learning for remaining useful lifetime prediction

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ABSTRACT

Accurate prediction of Remaining Useful Life (RUL) is essential for enhancing industrial reliability, reducing operational costs, and preventing unplanned downtime. However, conventional machine learning and deep learning approaches often suffer from overfitting, limited interpretability, and high computational overhead, making them less effective in dynamic and highly regulated predictive maintenance settings. To overcome these limitations, we propose the Quantum Inverted Transformer (QiTransformer), a novel architecture that integrates quantum-enhanced representation learning with causal inference and explainable artificial intelligence. In this framework, LiNGAM (Linear Non-Gaussian Acyclic Model) is employed to identify causative variables through a directed acyclic graph, enabling a more compact and computationally efficient feature space. The quantum components of the QiTransformer further strengthen its ability to extract temporal dependencies, resulting in more accurate RUL estimation. We evaluate the QiTransformer on two distinct predictive maintenance benchmarks: a battery degradation dataset and the NASA C-MAPSS aircraft engine dataset. Across both datasets, the proposed model consistently outperforms traditional deep learning and classical transformer-based approaches in terms of predictive accuracy and computational efficiency. Explainability analyses indicate that *Discharge Time (s)* is the most influential factor in the battery dataset, while *Corrected Core Speed (rpm)* emerges as the dominant feature in the C-MAPSS dataset. These results demonstrate the potential of quantum-enhanced, causality-aware architectures for next-generation predictive maintenance solutions.

1. Introduction

Predictive maintenance (PdM) plays a central role in modern industrial operations and uses the capabilities of Industry 4.0 and Industry 5.0. [1]. Unlike conventional maintenance policies that schedule interventions at fixed time intervals, PdM relies on continuous data analysis so that maintenance actions can follow the actual operating condition and behaviour of equipment in complex production settings. By monitoring asset health in real time, PdM supports early discovery of incipient faults, which in turn reduces unplanned outages and improves overall equipment availability [2]. This forward-looking strategy typically lowers maintenance costs and contributes to higher operational efficiency [3]. Moreover, real-time condition information allows organizations to allocate maintenance resources more selectively, focusing on assets that truly require attention [4,5]. As a result, PdM enhances the stability of industrial systems and diminishes the frequency of sudden operational disturbances [6].

A central capability in PdM is the estimation of RUL, defined as the time span over which a component can be expected to function reliably before intervention becomes necessary [7–9]. Over roughly the last decade, numerous modelling approaches have been proposed to characterize degradation behaviour and infer wear progression from streaming sensor data [10]. These methods have enabled detailed RUL forecasts that deliver tangible benefits in application areas such as manufacturing, transport, and energy infrastructures.

Data-driven approaches based on machine learning (ML) and deep learning (DL) have been particularly influential in advancing RUL prediction. Classical ML models, including gradient boosting, random forests, and support vector machines, often yield strong results when the input is well-structured and sufficiently large labeled datasets are available [11]. Deep architectures, especially recurrent neural networks (RNNs) and convolutional neural networks (CNNs), further extend these capabilities by capturing nonlinear effects and long-range temporal dependencies that are typical of industrial sensor signals [12].

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Nevertheless, these families of models present several important shortcomings.

A major issue is the "black-box" character of many ML and DL methods, whose internal reasoning is rarely transparent. This opacity is problematic in contexts where model interpretability is essential for adoption, such as healthcare, aviation, autonomous systems, and financial services [13]. Limited transparency makes it difficult to pinpoint which features drive predictions, to uncover biases or systematic errors, and to demonstrate conformity with regulatory requirements [14]. In addition, high-performing models frequently depend on large volumes of labeled data, which may be scarce or expensive to generate in industrial environments, and their accuracy can deteriorate when operating conditions deviate from those observed during training. To address these issues, this work proposes an integrated framework that combines causal inference, explainable artificial intelligence (XAI) [15], and a quantum inverted transformer (QiTransformer) architecture. Standard transformer architectures are already well regarded for their ability to capture long-range dependencies, but embedding quantum computational concepts can increase efficiency and expressive power when working with high-dimensional feature spaces. Quantum-inspired operations facilitate parallel exploration of complex input representations, offering potential computational advantages and improved scalability. The inverted design encourages backward reasoning from observed degradation states, which helps reconstruct degradation trajectories more precisely and supports more accurate RUL forecasts. Thus, the proposed framework yields a predictive maintenance approach that is causally grounded, interpretable, and computationally advanced. The salient contributions of this study are as follows:

- Design and assessment of a quantum enhanced inverted transformer capable of processing large scale industrial datasets and uncovering deep, nonlinear patterns relevant to RUL prediction.
- Integration of causal inference techniques to uncover credible cause and effect relationships among operational variables, thereby supporting trustworthy maintenance decision-making.
- Embedding of XAI mechanisms to ensure that the predictions of QiTransformer remain interpretable and practically useful for maintenance professionals.

The remainder of this paper is organized as follows. Section 2 provides an overview of related work. Section 3 details the proposed causal-inference-driven, XAI-supported quantum deep learning model for RUL prediction. Section 4 presents two case studies using benchmark datasets to evaluate the effectiveness of the framework, followed by concluding remarks in Section 5.

2. Related works

This section reviews existing works on PdM (Section 2.1) and RUL prediction (Section 2.2). These studies employ a variety of techniques including deep learning, transformers, GANs, meta-learning, and quantum inspired methods.

2.1. Existing works on predictive maintanance

A substantial portion of predictive maintenance research relies on deep learning architectures that analyze multivariate sensor streams to identify degradation patterns and forecast system health. Shoorkand et al. [16] proposed a dynamic model which integrates production planning with predictive maintenance (PdM) planning. They implemented a Rolling Horizon Planning approach to address the dynamic nature of production systems and introduced constraints that account for imperfect maintenance actions to enhance the model performance. They adopted a novel hybrid DL technique which combines CNN and LSTM networks. However, the complexity of combining these techniques may limit real time application feasibility. Dangut et al. [17] presented a novel DL technique combining auto encoder and bidirectional GRU networks for rare failure predictions in aircraft PdM. However the complexity and computational demand of the method may hinder its real-time applicability and scalability in large-scale operational environments. Qiu et al. [18] introduced the hybrid method which combines adaptive degradation stage division with temporal convolutional networks to train degradation stages. The complexity and reliance of the model on accurate stage division may limit its effectiveness in real time scenarios.

Another line of work examines quantum or quantum-inspired modeling approaches that aim to improve representation capacity and capture complex degradation behavior in predictive maintenance and battery health applications. Ghosh et al. [19] proposed an evolving technique for state of health (SoH) estimation of batteries under dynamic environments, using a quantum fuzzy network and LSTM based voltage forecasting. However the model accuracy may be limited by data quality, irregular discharge patterns, and model complexity. Gao et al. [20] proposed a data augmentation approach using quantum assimilation to estimate the SoH of lithium ion batteries with unusual degradation patterns. However the accuracy of the method may decline if early abnormal data are sparse or highly noisy. Yacoubi et al. [21] introduced a quantum based seahorse optimizer for association rule mining. The integration of quantum principles into the traditional algorithm leads to enhance exploration and handles multiple objectives effectively. However the computational overhead remains a challenge. Oh and Lee [22] proposed a quantum mechanics-based framework for correcting missing values in predictive maintenance data using stochastic differential equations and lemma. The method captures fine grained noise and variability effectively than traditional estimators. However, it does not provide explainability in its correction process. Maior et al. [23] introduced a quantum machine learning approach for diagnosing rolling bearing faults using hybrid quantum classical models with parameterized quantum circuits. The method outperformed a classical neural baseline across two vibration signal datasets. Despite promising gains in efficiency and accuracy the approach lacks explainability and remains constrained by current quantum hardware limitations.

Recent studies also investigate explainable AI techniques to increase transparency in predictive maintenance models and provide interpretable insights for operational decision making. Gawde et al. [24] presented an XAI based predictive maintenance method for rotating industrial machines. However the complexity of an algorithm may limit its practical implementation and scalability. Upasane et al. [25] proposed a type 2 explainable fuzzy for PdM in the water pumping industry. It excels in predicting equipment failures and interpretability to outperform traditional models. However the complexity can be challenging for average users to potentially limiting its widespread adoption. Choi et al. [26] proposed a real time explainable based anomaly detection architecture for PdM in manufacturing systems. The framework integrates various anomaly detection algorithms and model interpretation techniques to predict and explain shutdowns. However the complexity may pose challenges for non expert users to potentially hindering practical application. Despite these advances, predictive maintenance models continue to face constraints in efficiency, causal understanding and transparent reasoning, which motivate the causal feature selection and explainability components of the QiTransformer.

2.2. Existing works on RUL prediction

A large body of RUL research applies deep learning based feature extraction methods that map high dimensional sensor signals into compact health indicators for degradation modeling. Liu et al. [27] proposed a multivariate phase warping method for tracking degradation and estimating the RUL of rolling bearings. This method requires extensive computational resources and detailed sensor data to potentially limiting its real time applicability and usefulness in environments with sparse data. Pan et al. [28] presented a meta weighted framework that includes distinguish in distribution based on out of distribution data for remaining lifetime prediction in turbopump bearings by estimation of uncertainty. This model adaptively reweights samples based on prediction loss but is hindered by the need for precise loss gradients which may complicate its practical implementation. Suh et al. [29] developed a multiscale feature extraction approach using generative adversarial networks specifically a UNet based GAN to capture sequence features from vibration signals for bearing RUL prediction. Although this approach captures detailed sequence features,it may struggle with efficiency and scalability when dealing with complex patterns and high dimensional data. Xu et al. [30] introduced a convolutional autoencoder networks in conjunction with a status degradation model to extract features and transform them into a health status index for RUL prediction. Although this approach is effective, its reliance on precise feature extraction could hinder its adaptability in real time applications. Xia et al. [31] introduced an ensemble method using a convolutional LSTM network with fault knowledge transfer which incorporates domain adaptation and fault information to handle various degradation patterns. However this reliance on extensive fault data could limit its applicability in diverse operational settings. Li et al. [32] proposed the deep multiscale feature for predicting the RUL of aero engines. This method combines extraction of multiscale feature with GRU based temporal learning. However the high data requirements might restrict its real time performance and generalization capabilities across different engines. Shang et al. [33] proposed an automated technique that combines Bi-GRU with CNN for rolling element bearing RUL prediction. However the generation of weighted averaging method lacks to diverse different operational conditions.

Another group of studies focuses on recurrent and attention based models that enhance temporal feature representation through gated units, dual attention mechanisms and probabilistic modeling frameworks. Shi et al. [34] proposed a LSTM with dual attention for exponential smoothing and splits features into aggregated encoding and original features. This model effectively predicts the RUL but its dual attention mechanism leads to risk of information loss during feature encoding. Chen et al. [35] developed an attention based RNN that extracts bandpass energy features to predict health index (HI) and then link these value to RUL through linear regression. This performance is heavily reliant on accurate feature extraction and limited to trained operational conditions. Qin et al. [36] presented the dual gated attention network which integrates GRU with dual attention mechanisms. However it may suffers from increased computational complexity and resource demands. Lin et al. [37] proposed a method using a multi nonlinear wiener and a variational bayesian approach to estimate the RUL of mechanical components. This process accounts for unit to unit variability and change point degradation. Zhou et al. [38] introduced the reinforced memory GRU model for RUL bearings estimation. The network reuses previous state information to mitigate forgetting issues in RNN. However the model lacks to overcome computational intensity.

A separate line of work examines domain adaptation and similarity based methods that aim to maintain predictive performance across heterogeneous operating conditions. Zhou et al. [39] developed an adaptive method for aeroengine RUL prediction using multi angle similarity evaluation. This approach uses a convolutional autoencoder to compute a HI and dynamic time warping for global similarity assessment. However the model mainly depends on the HI such that it lacks to generalize the model for distinct types of engines. Ding et al. [40] presented the multisource domain adaptation network (MDAN) for predicting bearing RUL on various operational conditions. MDAN enhances generalization by extracting domain invariant features from time frequency vibration signals and its dependence on diverse source domains may restrict to single domain scenarios. Yang et al. [41] introduced the Meta DESK approach which utilizes a deep sparse kernel network using meta learning for few shot RUL prediction. This model incorporates gaussian process and variational inference within a meta learning framework to manage sparse data and reduce overfitting. However the requirements of the model for extensive tuning may limit its practical applications especially in environments where simplicity and fast deployment are crucial.

Recent contributions explore transformer architectures, reinforcement learning frameworks and quantum inspired models that seek to improve long range dependency modeling and computational efficiency in RUL prediction. Chen et al. [42] developed a transformer framework to estimate the RUL of lithium ion batteries to capture temporal features and integrating denoising with prediction tasks. This approach offers detailed temporal feature capture but may be hindered by its complexity and high computational demands. Zheng et al. [43] proposed a novel teachnique for estimating the RUL using deep reinforcement learning. This technique combines feature extraction through an autoencoder with reinforcement learning to preserve temporal correlation in the data. Despite its potential for sophisticated feature extraction, the complexity and dependence of the method on accurate feature extraction could restrict its practical applicability. Liu et al. [44] proposed an architecture for estimating the RUL with quantum neural networks and dynamic complexity characteristic entropy. They introduced a novel entropy measure incorporating energy density changes to better identify fault patterns and design a principal component analysis based variational autoencoder to generate an effective HI. However, the performance of the method may be sensitive to the complexity of quantum network optimization and lacks explainability.

Estimating RUL draws on a diverse set of analytical methods, each offering distinct advantages while also presenting practical limitations (see Table 1). Conventional techniques place heavy emphasis on carefully engineered features, which often demand considerable computational effort and tend to struggle when available data are limited. More recent models based on recurrent neural networks and attention layers capture temporal behaviour effectively, yet this sophistication comes with added computational cost. The Quantum Inverted Transformer (QiTransformer) proposed in this study offers a new direction for RUL estimation by bringing together causal reasoning, quantum-inspired attention, and model interpretability within a single framework.

3. Proposed methodology

The proposed quantum inverted transformer (QiTransformer) presents a novel framework for RUL prediction by integrating causal inference with quantum attention and explainability. Unlike existing quantum transformers the proposed model establishes a formulation that achieves interpretability and training efficiency. Causal pruning based on the linear non gaussian acyclic model (LiNGAM) retains only those variables that exert direct causal influence thereby reducing input dimensionality for quantum embedding. Attention computation is reformulated using parameterized quantum circuits that derive weights from quantum correlation as expectation values. This quantum attention captures temporal dependencies that classical self attention may not represent effectively. An integrated explainability framework combining SHAP and LIME with counterfactual analysis which further interprets the influence of causal variable on the predicted RUL by connecting quantum modelling with transparent decision support for maintenance planning. The complete framework of the proposed methodology is presented in Fig. 1.

3.1. Causal inference

Causal inference is critical in predictive maintenance to identify the underlying relationships within datasets. We adopted the LiNGAM [47] method for identifying causal variables in systems where the relationships are linear and are not gaussian. The key assumptions of the LiNGAM for causal discovery are as follows.

- Acyclicity: The causal graph is directed and acyclic which means no directed cycles or feedback loops exist.
- Independence: Exogenous variables are independent which implying no latent confounders influence multiple observed variables.

Table 1
Summary of existing models.

Reference	Model(s)	Objectives	Remarks
[27]	Multivariate Phase Space Warping	Track degradation and predict RUL of rolling bearings	Requires extensive computational resources and detailed sensor data.
[28]	Meta-weighted neural network	Adaptively reweight samples from out of distribution.	Complexity and reliance on accurate loss gradients.
[34]	Dual Attention LSTM	Predict RUL by capturing aggregated encoding and original features	This model may suffer from computational overhead and risks information loss during encoding.
[45]	Multiscale Hourglass Transformer	Predict RUL of aircraft engines with multi-scale feature extraction	The model needs huge computational resources.
[39]	Convolutional autoencoder	To predict RUL of aero engines	The model may limit efficiency and generalizability.
[41]	Meta learning and deep sparse network	To perform Few-shot RUL prediction	High complexity and extensive tuning requirements.
[37]	Variational Bayesian approach	Predict RUL considering uncertainty and unit-to-unit heterogeneity	Complex Bayesian updates and nonlinear modeling pose computational challenges.
[40]	Multisource Domain Adaptation Network	RUL prediction of bearings across various operating conditions.	Complexity and dependence on diverse source domains may limit applicability.
[29]	Generative Adversarial Networks with U-Net	To estimate RUL of bearings.	Lacks efficiency and scalability compared to Quantum Deep Learning (QDL) in handling high-dimensional data.
[35]	Attention based recurrent neural network	To predict RUL of bearings	Reliance on accurate feature extraction and linear regression.
[42]	Transformer-based network	To estimate RUL of lithium ion batteries.	High computational demand and complexity and lack of explainability.
[32]	Integrated Deep Multiscale Feature Fusion Network	To perform RUL prediction of aero engines.	Complexity and high data requirements may limit generalize across engines.
[31]	Convolutional LSTM network	RUL prediction by learning degradation patterns under different faults via fault knowledge transfer	Complexity and reliance on extensive fault data.
[36]	Gated Dual Attention Unit Neural Network	Predict RUL of rolling bearings with gated recurrent units and dual attention mechanisms	Increased resource demands may limit its practical deployment.
[38]	Reinforced Memory Gated Recurrent Unit Network	Predict RUL of bearings by reusing state information to overcome rapid forgetting in RNNs	Lacks to provide explainability analysis.
[18]	Temporal Convolutional Network	Predict RUL of bearings by modeling degradation in stages with TCNs	Accurate stage division may limit real time effectiveness and adaptability.
[30]	Convolutional Autoencoder	Predict RUL of rolling bearings by extracting features and mapping health status	Accurate feature extraction and modeling lmay leads to lack of adaptability and effectiveness.
[46]	Cost-Sensitive Learning with Deep Feature Transfer	Estimate RUL of bearings using imbalanced regression	Reliance on accurate discretization and deep feature transfer.
[33]	Bidirectional Gate Recurrent Unit	Predict RUL for elements of rolling bearings.	The models lacks to analyze diverse operational conditions.
[43]	Deep Reinforcement Learning	Predict RUL of rolling bearings using reinforcement learning.	The model suffers to perform accurate feature extraction.
[26]	Explainable anamoly detection framework	To perfrom predictive maintenance in manufacturing systems.	The model require lot of expertise.

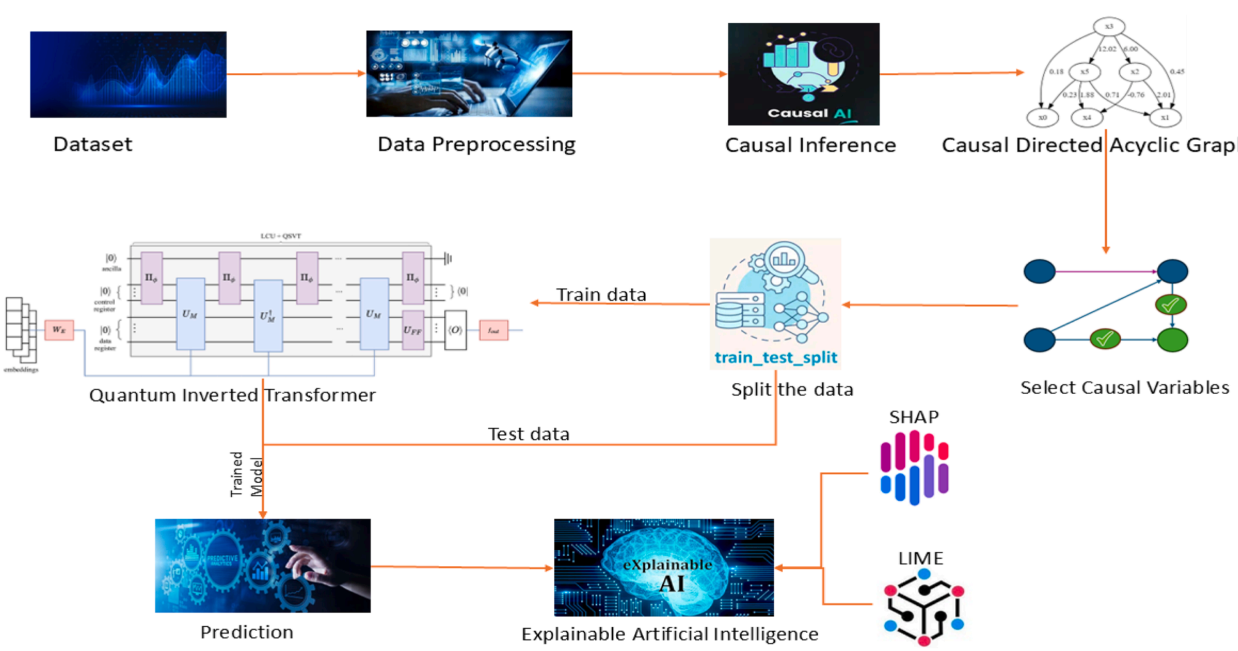


Fig. 1. Framework for proposed methodology.

- Linear Relations: Assumes linear functional relationships among variables.

For continuous observed variables x_i (where $i = 1, \dots, d$), the model is defined as follows (see Eq. (1)).

$$x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i \tag{1}$$

where e_i are continuous, independent, and non-Gaussian exogenous variables. The matrix form of this is represented in Eq. (2).

$$x = Bx + e \tag{2}$$

where, B is the connection strength matrix, and x and e are vectors of observed and exogenous variables, respectively. The absence or presence of a edge directed from x_j to x_i is determined by whether $b_{ij} = 0$. The connection strength of the matrix B can be estimated by using independent component analysis (ICA) [48] and x can be transformed into an ICA model which are represented in Eq. (3).

$$x = Ae \tag{3}$$

where $A = (I - B)^{-1}$. This approach leverages ICA to estimate $W = A^{-1}$ which leads to the determination of B is presented in Eq. (4).

$$B = I - W \tag{4}$$

The simplified LiNGAM for causal discovery is presented in Eqs. (5)–(7).

$$x_1 = e_1 \tag{5}$$

$$x_2 = b_{21} x_1 + e_2 \tag{6}$$

$$x_3 = b_{32} x_2 + e_3 \tag{7}$$

where e_1,e_2 and e_3 are non gaussian and mutually independent. To integrate causal discovery into the QiTransformer, the connection strength matrix B is used to identify variables with direct causal influence on degradation. A variable is retained if at least one coefficient b_{ij} associated with its incoming edges exceeds a predefined threshold. Variables that do not satisfy this condition are removed from the input space. This pruning step yields a reduced feature subset, denoted X_{causal}, which contains only the causal drivers of RUL.

The causal pruning step reduces the dimensionality of the sensor data from the full set of d variables to a smaller subset d_{causal} [49]. This reduction improves the efficiency of the quantum embedding stage because the number of qubits and rotation parameters depends on the input dimension. A smaller feature set also lowers the depth of the entanglement operations and reduces the cost of gradient evaluation during hybrid optimization. The reduced causal feature set X_{causal} forms the input to the quantum inverted transformer block in Section 3.2.

3.2. Quantum inverted transformer model

We propose a quantum inverted transformer (QiTransformer) model (see Fig. 2) which leverages quantum computing principles within an inverted transformer architecture [50] to maximize the RUL prediction. The model combines classical components (such as the transformer architecture) with quantum elements (such as quantum embedding and quantum operations) making it a hybrid quantum classical model. The QiTransformer focuses on encoding the variates into quantum states to process them using quantum operations and generating future predictions through a combination of quantum and classical layers. In the context of estimating the RUL of a system we have a dataset $X = x_1, x_2, \dots, x_T$ where each x_t represents a collection of measurements taken at a certain point in time. These measurements include N different attributes or variates capturing various aspects of the systems condition. Our goal is to forecast the behavior of the system for the S time steps which represented as $Y = x_{T+1}, x_{T+2}, \dots, x_{T+S}$. This prediction helps estimate how much longer the system can operate effectively before requiring maintenance or facing potential failure.

Given observations $X = x_1, \dots, x_T \in R^{T*N}$ with T tuples and N variates then we predict the future S time steps $Y = x_{T+1}, \dots, x_{T+S} \in R^{S*N}$. X_t denotes the collection of all variates at time step t and X_n represents the time series of each variate indexed by n. It is significant that X_t may not reflect simultaneous real world events due to systematic time lags among variates, and each element of X_t can differ in physical measurement. The key architectural components of QiTransformer are described as follows.

Quantum Embedding: The embedding layer receives the causal feature set X_{causal} identified from Section 3.1. Each variable is encoded into a quantum state through angle encoding using rotation gates $R_y(x_i)$ and $R_z(x_i)$. The size reduction obtained from causal pruning decreases the number of qubits and rotation parameters, which improves the efficiency of the embedding layer.

- Each variate series X_n is embedded into a high dimensional quantum state using a parameterized quantum circuit. This embedding maps the classical bits into the quantum domain by capturing the unique properties of each sensor reading over time.
- The embedding $R^T \rightarrow R^D$ is implemented using parameterized quantum circuits (PQCs) via angle encoding that encode the variate series into a D dimensional quantum state where each feature value is mapped to a quantum rotation gate $R_y(\theta)$ or $R_z(\phi)$. The PQC layer uses $n_{qubits} = 3$ qubits arranged in a linear topology with entangling `CZ` gates applied in a strong entanglement pattern across adjacent qubits. Each embedding layer includes two parameterized rotation layers (Ry, Rz) followed by the entanglement layer thereby yielding 6 trainable parameters per layer. The total quantum parameters scale as $O(n_{qubits} \times L_{PQC})$ where L_{PQC} is the PQC depth.

Quantum Inverted Transformer Blocks: The QiTransformer replaces classical dot product attention with quantum attention derived from expectation values. For parameterized quantum states $|\psi(\theta)\rangle$, the attention score between positions i and j is computed as $\alpha_{ij} = \langle \psi(\theta_i)|O|\psi(\theta_j)\rangle$. This formulation captures correlations among the causal variables that classical attention cannot represent effectively.

- The quantum encoded variate tokens $H_0^n = \text{Embedding}(X_n)$ are processed using a stack of L quantum inverted transformer blocks.
- Each block contains quantum attention mechanisms and quantum feed forward networks which are designed to capture the interdependencies and correlations between different sensor measurements.
- Quantum attention mechanisms utilize quantum operations such as entanglement and superposition to serve as attention weights to model the interactions among variates. This is denoted as $H_{l+1} = \text{QuantumITrmBlock}(H_l), l = 0, \dots, L-1$.

The reduced feature dimension d_{causal} lowers the depth of the entanglement layers and reduces the computational cost of hybrid optimization. This effect is important because the number of controlled Z gates and the cost of gradient evaluation both scale with the input dimension.

Projection: The output of the quantum attention block is processed by a classical projection layer that maps the learned quantum representations to the final RUL estimate.

- After processing through the quantum inverted transformer blocks and the final quantum H_L^n state is projected back into the classical domain to generate the future predictions of the RUL.
- The projection process Projection:$R^D -> R^S$ is implemented using classical layers that transform the quantum-processed features into the predicted readings are represented as $\hat{Y}_n = \text{Projection}(H_L^n)$. The

hybrid output layer combines quantum measurement features with classical dense layers to generate RUL estimates with reduced variance.

Here, $H = \{h_1, \dots, h_N\} \in \mathbb{R}^{N\times D}$ represents the quantum-encoded variate representations. Both embedding and projection are implemented as hybrid quantum-classical mappings to enable end-to-end differentiability.

Quantum Layer Normalization:
Traditional layer normalization stabilizes training and improves convergence by normalizing features across time steps. In typical Transformer-based RUL forecasters, this normalization is applied to multivariate representations of the same timestamp. However, this approach can introduce noise from non-causal interactions if time points do not represent the same event due to time lags or measurement discrepancies. In the QiTransformer the layer normalization is adapted to handle the quantum states of individual variate tokens [51] which helps to manage non stationary problems and normalize the data to a gaussian distribution to reduce inconsistent measurements. This can be formulated as follows (see Eq. (8))

$$\mathrm{LayerNorm}(H) = \frac{h_n - \mathrm{Mean}(h_n)}{\sqrt{\mathrm{Var}(h_n)}}, \quad \text{for } n = 1, \dots, N \tag{8}$$

Quantum Feed Forward Networks
In classical Transformers the feed forward network (FFN) encodes token representations and is applied identically to each token. This can be localized and failed to capture sufficient predictive information on dealing with multivariate sensor data for RUL prediction. In the QiTransformer, the FFN is applied to the series of each variate token to leverage the universal approximation to extract the complex sensor data over time. Stacked quantum inverted transformer blocks encode the observed variate data and decode them for future predictions utilizing dense non linear connections. These connections effectively encode intrinsic properties of the sensor data, such as trends, periodicities, and anomalies.

Quantum Self-Attention
Self-attention in traditional models facilitates temporal dependency modeling. In the QiTransformer, each sensor's data series is treated as an independent process. With representations $H = h_0, \dots, h_N \in R^{N*D}$, the self-attention mechanism uses quantum operations to project these into queries, keys, and values $Q, K, V \in R^{N*d_k}$ is the projected dimension. The pre-softmax scores are computed as follows (see Eq. (9)).

$$A_{i,j} = \frac{(QK^T)_{i,j}}{\sqrt{d_k}} \alpha q_i^T k_j \tag{9}$$

Once each token is normalized on its dimension of feature then the score map $A \in R^{N*N}$ reveals the correlations between different sensor readings. This enables the model to weigh highly correlated sensors more heavily for the next representation interaction with values V. Expectation values from PQCs are used as entries in Q and K, linking quantum correlations to attention scores.

Computational Complexity Analysis:

- The classical Inverted Transformer has complexity $O(N^2 \cdot d_k)$ for attention and $O(N \cdot D^2)$ for feed-forward layers.
- For QiTransformer, quantum embedding adds $O(N \cdot (G \times n_{qubits}))$ operations, where G is the number of quantum gates per PQC. The hybrid model reduces total complexity because quantum embeddings reduce the effective feature dimension D.
- With $n_{qubits} = 3$ and shallow PQCs (depth = 2), simulation is efficient on classical backends. The total runtime scales linearly with the number of tokens, $O(N)$, under current configurations.

Parameterization Strategy: Model parameters include classical transformer weights and quantum circuit angles θ_i. Gradients for both are computed using the parameter shift rule allowing hybrid training via backpropagation. Quantum layers are implemented using the `default.qubit` simulator from the PennyLane framework which ensures reproducibility across hardware and simulators.

Table 2
Hyperparameters for QiTransformer.

Parameter	Value
Seq_len	96
Dropout	0.1
Number of qubits (n_qubits)	3
Activation function	ReLU
Quantum backend	`default.qubit`
Learning rate	1×10^{-3}
Embedding type	Angle
Attention heads	4
Weight decay	0.01
Gradient clipping	1.0
Optimizer	Adam
Class strategy	Projection
Entanglement type	Strong
Epochs	250
Self-attention dim (d_k)	32

The hyperparameters utilized to implement the proposed model are enumerated in Table 2.

This simulation is implemented using the *default.qubit* backend used to reproduces quantum state transformations and expectation value computations efficiently on classical hardware. This allows the proposed model to use representational learning advantages of quantum principles without physical quantum processors. Further the architecture is hardware agnostic such that which can be executed on actual quantum devices once available for future quantum acceleration. The following elements distinguish the proposed QiTransformer from existing transformer-based models.

- Attention scores are computed from quantum correlations expressed as expectation values.
- The quantum attention mechanism uses the causal subset X_{causal} instead of the full input space.
- Controlled Z entanglement captures nonlinear dependencies among causal variables.
- Hybrid optimization with the parameter shift rule updates the quantum circuit efficiently.

3.3. Explainable quantum inverted transformer model

The explainability module interprets how the causal variables and the quantum attention mechanism influence the predicted RUL. This module evaluates the contribution of each causal variable in X_{causal} and provides transparent decision support for maintenance planning. Transformer models have superior predictive accuracy compared to simpler ML and DL models. Despite their effectiveness, these models often represented as blackbox in nature which means that their internal mechanisms are complex to interpretable by humans. Explainability in ML refers to clarify how the predictions are arrived [52,53]. This capability is significant for making informed decisions in scenarios such as predicting the RUL of systems where understanding the cause behind predictions is essential to implement targeted interventions [54]. Quantum attention produces correlation patterns that are not directly observable, so an explainability layers required to interpret which causal variables influence the model output and how these influences vary across time.

The interpretability of ML models can be performed using two main perspectives such as global and local. Global interpretability gives a comprehensive of how a model makes estimations by examining its entire feature set and aims to explain at a high level of abstraction. In contrast, the local interpretability focuses on explaining individual predictions to provide insight into why a model arrive a specific conclusion for a given input instance. The explainability analysis is applied only to the causal subset X_{causal} rather than the full feature space. This alignment ensures that the explanations correspond to variables with direct

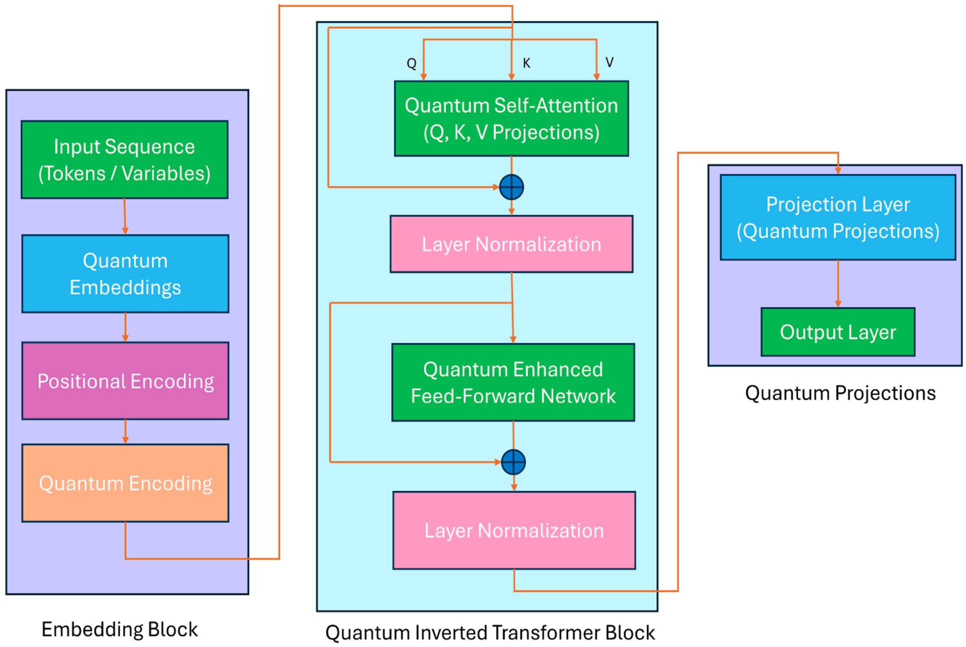


Fig. 2. Architecture of Quantum Inverted Transformer Model.

influence on degradation and avoids attribution to irrelevant features. In our research context, we employ SHAP [55], a model agnostic approach rooted in game theory. SHAP calculates the contribution of each feature using Shapley values. The Shapley index for feature i which denoted as $\phi_i(v)$ is calculated using the following Eq. (10).

$$\phi_i(v) = \sum_{S \subseteq F\{i\}} \frac{|S|! \cdot (|F| - |S| - 1)!}{|F|!} \cdot [v(S \cup \{i\}) - v(S)] \tag{10}$$

where v represents the total model output, F denotes the entire feature set, S is a features subset that exclude feature i, $|S|$ is the number of elements in subset S, and $v(S \cup \{i\}) - v(S)$ measures the incremental impact of including feature i in subset S. This formula encapsulates how SHAP assigns importance to each feature considering all possible combinations. SHAP values quantify the marginal contribution of each causal variable to the predicted RUL, which enables the identification of degradation drivers that consistently influence the quantum attention output. Local explanations from LIME complement the global attributions from SHAP by identifying the causal variables that influence individual predictions. Counterfactual analysis evaluates how small perturbations in selected causal variables alter the predicted RUL, which provides additional insight into degradation sensitivity.

The interpretability of proposed model is inspirational due to their hybrid nature. The quantum processing steps involve transformations that are not directly interpretable. SHAP enables us to dissect how Qi-Transformer and classical feature interactions influence the model predictions. This approach helps to reconcile the gap between quantum enhanced computations and classical interpretability by giving insight into feature significance. The integration of causal modeling with the QiTransformer and the explainability analysis are represented in the Algorithm 1.

The combination of SHAP, LIME, and counterfactual evaluation provides a coherent explanation of the influence of each causal variable on the quantum attention output. This integration produces transparent and reliable diagnostic information that can support maintenance engineers in understanding degradation patterns and adjusting inspection or replacement schedules.

Algorithm 1 Causal Inference and Explainable Artificial Intelligence for Quantum Inverted Transformer.

Input: 'n' number of predictive maintenance variables.

Output: Identify causal variables, analyze the patterns to predict the RUL and provide explanations using XAI.

1: Choose the variables which impact the life cycle of the systems under consideration.
2: Employ the LiNGAM algorithm to discover the causal structure among the variables.
3: **for** continuous observed variables x_i (where $i = 1, \dots, d$), then the model **do**
  $x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i$
4: **end for**
5: The simplified LiNGAM model is applied to identify the caused variables using ICA approach.
  $x_2 = b_{21} x_1 + e_2$
  $x_3 = b_{32} x_2 + e_3$
  $x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i$
6: Compute the causal adjacency matrix representing the strength and direction of causal relationships.
7: Initialize the Quantum Inverted Transformer model.
8: Each variate series X_n is embedded into a high-dimensional quantum state using a quantum circuit.
9: The Quantum Inverted Transformer model involves
  $R^T \to R^D$
  $H_0^n = Embedding(X_n)$
  $H_{l+1} = QuantumITransBlock(H_l), l = 0, \dots, L-1$
  $\hat{Y}_n = Projection(H_L^n)$
10: Predict RUL using the Quantum Inverted Transformer on causal variables.
11: Use explainable SHAP and LIME plots to interpret the predictions of the model.
12: Compute feature importance to understand participation of each feature to the decisions of the model.
13: Generate summary and dependence plots to visualize feature impact and interactions.
14: Provide Explanation insights and visualizations.

Table 3
Descriptive Statistics of the Predictive Maintenance Dataset.

Feature	count	mean	std	min	25%	50%	75%	max
Cycle_Index	15,064	556.15	322.37	1.0	271.00	560.00	833.00	1134.00
Discharge Time (s)	15,064	4581.27	33144.01	8.69	1169.31	1557.25	1908.00	958320.30
Decrement 3.6-3.4V (s)	15,064	1239.78	15039.58	-397645.90	319.60	439.23	600.00	406703.768
Max. Voltage Dischar. (V)	15,064	3.90	0.09	3.04	3.84	3.90	3.97	4.36
Min. Voltage Charg. (V)	15,064	3.57	0.12	3.02	3.48	3.57	3.66	4.37
Time at 4.15V (S)	15,064	3786.33	9129.55	-113.58	1828.88	2930.23	4088.32	245101.11
Time constant current (s)	15,064	5461.26	25155.84	5.98	2564.31	3824.26	5012.35	880728.10
Charging time (s)	15,064	10066.49	26415.35	5.98	7841.92	8320.41	8763.28	880728.10
RUL	15,064	554.19	322.43	0.000	277.00	551.00	839.00	1133.00

4. Results and discussions

In our work we employed two distinct benchmark datasets to predict the RUL of machinery. The first dataset comprises data with batteries capacity of $2.8Ah$ which are cycled over 1000 times at 25 degree celsius with a $CC-CV$ charge rate of $C/2$ rate and discharge rate of $1.5C$ by capturing behavior of current and voltage of each cycle of the batteries. This dataset enables us to monitor the operational health of the devices over time and estimate when maintenance could be performed to preemptively address breakdowns. Complementing this we utilize the NASA C-MAPSS (Commercial Modular Aero Propulsion System Simulation) dataset which provides comprehensive life cycle data for aircraft engines. This dataset is instrumental in modeling and estimating the RUL of aircraft engines providing detailed insights into the operational parameters and degradation patterns over their entire lifespan. By integrating insights from both datasets we aim to develop robust predictive maintenance of the model that enhance reliability and efficiency by extending the operational life of the machinery and reducing downtime.

4.1. Experiments and analysis on predictive maintenance dataset

The study draws on a publicly available predictive-maintenance dataset hosted on Kaggle (https://www.kaggle.com/datasets/ ignaciovinua les/battery-remaining-useful-life-rul), which contains detailed operational records for lithium-ion batteries. The dataset provides nine key attributes that describe how each battery behaves over its life cycle. *Cycle Index* represents the discharge cycles of the battery where a key indicator of its usage history. *Discharge Time (s)* measures the duration the battery operates before recharging. *Time at 4.15V (s)* indicates what extend the battery maintains a voltage level during discharge. *Time Constant Current (s)* reflects the duration over which a constant current is maintained during discharge. *Decrement 3.6-3.4V (s)* measures the time taken for the voltage to decrease from 3.6V to 3.4V during discharge characteristics. *Max Voltage Discharge (V)* denotes the highest voltage output during discharge while *Min Voltage Charge (V)* signifies the lowest voltage level when the battery is recharged. *Charging Time (s)* quantifies how long it needs to recharge the battery fully. Total time represents the cumulative operational time of the battery while *RUL* predicts the expected duration before the performance of the battery falls below acceptable levels. These variables collectively capture the dynamic operational behavior of the battery degradation trends. Consequently this dataset effectively represents a dynamic PdM environment where non stationary operating conditions and performance variations over time influence RUL estimation. The summary of the PdM dataset is represented in Table 3 and the summary of variables for the PdM dataset is provided in Table 4.

4.1.1. Data preprocessing

Data preprocessing began with eliminating missing entries to prevent distortions in model training. We also examined the dataset for outliers, applying statistical techniques to flag values that deviated markedly from normal operating patterns and could signal atypical device behaviour. Leaving such irregularities untreated may bias the RUL predictions-either by implying an early failure or by masking genuine degradation trends. Through this systematic cleaning process, we ensured that the dataset provided a reliable basis for constructing and evaluating the PdM models.

4.1.2. Causal inference on predictive maintenance dataset

We employed causal inference to uncover the underlying relationships between the variables. Unlike traditional correlations that capture associations, we used LiNGAM to identify causality among the nine variables listed in Section 4.1.1. LiNGAM is effective for detecting causal relationships where the data is non-Gaussian and the structure is acyclic. By applying LiNGAM, we could analyze changes the variables that causally affect battery failures. The causal directed acyclic graph is presented as follows (see Fig. 3).

Using the LiNGAM we identified several key variables that have a direct causal impact on the RUL. The analysis revealed that *Discharge Time (s)*, *Min. Voltage Charg (V)*, *Charging Time (s)*, *Time constant current (s)*, *Time at 4.15 (V)*, and *Decrement 3.6-3.4V (s)* are contributors to the RUL. These variables are crucial as they provide operational context and captures the degradation patterns. By focusing on these causal variables for our model training we streamlined the feature set and thereby optimizing the computational resources. This targeted approach reduced the complexity without compromising its predictive accuracy. Instead of training on all variables which could include redundant data. Hence we concentrated only on the factors that directly influence the RUL. This strategy can accelerated the training process and the interpretability by making it easier to deploy in real world maintenance applications.

4.1.3. Actual vs predicted values on predictive maintenance dataset

To predict the RUL of the devices we evaluated advanced ML models tailored for time series forecasting. Traditional LSTM networks result robust performance in capturing temporal dependencies by providing reliable RUL predictions. Quantum LSTM models leveraged quantum computing principles to enhance accuracy by exploring complex patterns within the data. Bi-LSTM networks including classical and quantum enhanced which consider past and future contexts simultaneously which refine RUL estimates by analyzing sequence data. GRU models including both classical and quantum enhanced are offered efficient architecture. Transformer models which are known for their sequence to sequence capabilities and their quantum counterparts showed exceptional performance in identifying long range dependencies. Moreover the QiTransformer emerged as the standout performer which showcasing superior performance and underscoring its capability to optimize predictive maintenance strategies. The actual versus predicted values for LSTM, Quantum LSTM, Bi-LSTM, Quantum Bi-LSTM, GRU , Quantum GRU, Inverted Transformer, Quantum Inverted Transformer are represented in Fig. 4a to g respectively.

4.1.4. Performance evaluation on predictive maintenance dataset

Evaluation parameters such as mean squared error (MSE), root mean squared error (RMSE), mean absolute percentage error (MAPE), mean absolute error (MAE), and computation time are used for analyzing the performance of predictive approaches. All models evaluated on this

Table 4
Summary of variables in the Predictive Maintenance (Battery) dataset.

Variable Name	Description	Variable Type	Environment Characteristic
Cycle_Index	Number of charge-discharge cycles	Continuous	Dynamic
Discharge Time (s)	Duration of battery discharge	Continuous	Dynamic
Decrement 3.6-3.4V (s)	Time for voltage drop from 3.6V to 3.4V	Continuous	Dynamic
Max. Voltage Dischar. (V)	Maximum voltage during discharge	Continuous	Dynamic
Min. Voltage Charg. (V)	Minimum voltage during charge	Continuous	Dynamic
Time at 4.15V (S)	Duration battery maintains 4.15V during discharge	Continuous	Dynamic
Time constant current (s)	Duration of constant current during discharge	Continuous	Dynamic
Charging time (s)	Duration of full battery recharge	Continuous	Dynamic
RUL	Remaining Useful Life (target variable)	Continuous	Dynamic

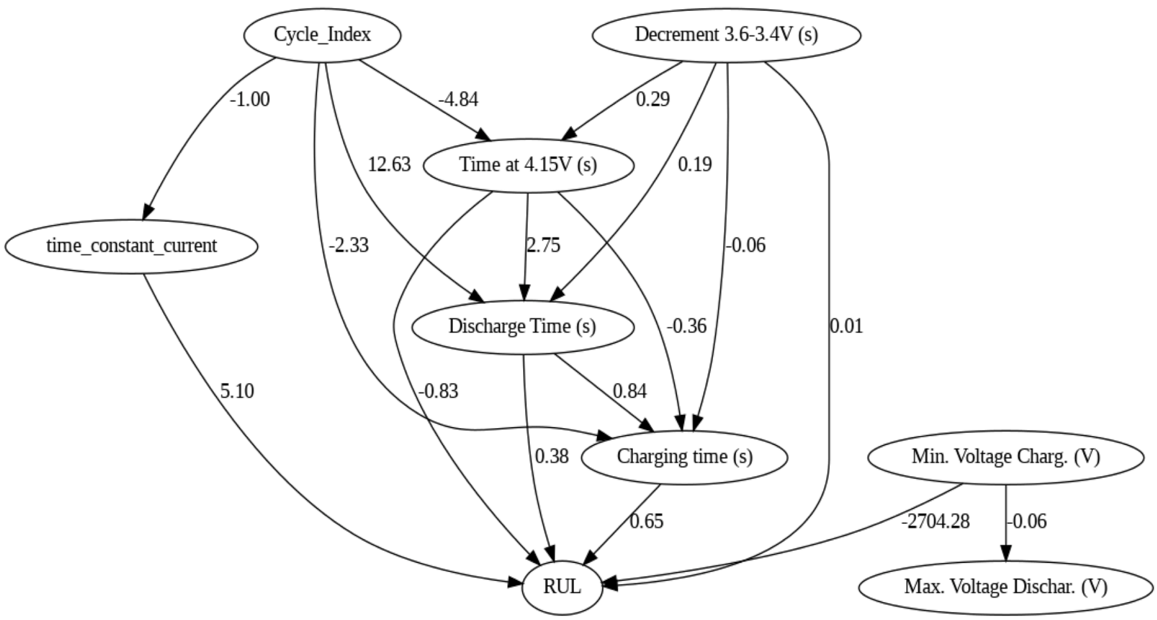


Fig. 3. Causal Directed Acyclic Graph (DAG) illustrating the causal relationships governing the Predictive Maintenance dataset.

dataset were re implemented using the same preprocessing pipeline, feature configuration, sequence construction, and train-test split as the QiTransformer to ensure a controlled and fair comparison. MSE given by MSE $= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ which measures the average squared variation between actual and predicted values could penalizing larger errors severely. RMSE represents the square root of MSE RMSE $= \sqrt{\text{MSE}}$ gives an error metric in the same unit as the target variable by making it more interpretable. MAE can defined as MAE $= \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$ which calculates the average absolute errors by providing a straightforward interpretation of accuracy. MAPE is given by MAPE $= \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$ which demonstrates the error as a percentage by providing prediction accuracy.

We conducted a thorough evaluation of both classical and quantum-enhanced models using key performance metrics such as MSE, RMSE, MAE, MAPE, and computation time. The traditional LSTM network [7] was employed as a baseline, yielding an MSE of 913.36 and a relatively high MAPE of 79.02. The Quantum LSTM model [56] demonstrated improved performance across all error metrics and faster computation, highlighting the benefits of integrating quantum principles for capturing complex data patterns. The DiffusionRUL method [57] showed strong predictive ability by leveraging diffusion-based generative modeling, achieving an MSE of 748.16 and MAPE of 62.44. Bi-LSTM networks [58], which process sequences bidirectionally, further reduced the MSE to 618.54. The Quantum Bi-LSTM [59] surpassed this with a notable MAPE reduction to 30.11, indicating enhanced modeling of bidirectional dependencies and complex temporal sequences. Graph Neural Networks (GNNs) [60] were explored to capture relational dependencies among sensor and operational variables, achieving an impressive MSE of 525.12 and MAE of 46.7 by effectively exploiting structural correlations often overlooked by sequence-only models. GRU networks [61] yielded commendable results with an MSE of 426.78, MAE of 53.18, and MAPE of 33.31, demonstrating their efficiency in prediction tasks. The Quantum GRU [62] further improved accuracy slightly, reducing MAPE to 33.01. The inverted Transformer model [50], known for adept sequence-to-sequence learning, achieved an MSE of 263.88 and RMSE of 16.24, with a respectable MAPE of 32.86. Notably, the proposed QiTransformer significantly outperformed all compared methods, attaining the lowest MSE of 153.98, RMSE of 12.41, MAE of 36.90, and an outstandingly low MAPE of 14.19, alongside efficient computation time of 31.87 s. Table 5 summarizes the comparative performance analysis against state of the art models.

4.1.5. Explainability analysis on predictive maintenance dataset

In our study on predicting the RUL that we harnessed the QiTransformer model and applied a diverse array of explainability to dissect the contributions of predictive variables. We employed beeswarm and beeswarm absolute plots to visualize the influence of each feature. Decision and bar charts provide comparisons of individual feature contributions while waterfall charts enabled us to quantify the cumulative effect of features. Cohort analysis was used to identify patterns within distinct subsets of the data which offering insights into feature behaviors. Further we utilized heatmaps and violin plots to display the distribution and density of predictions across the dataset. The LIME was employed for localized insights into the decision making. This multidimensional approach allowed us to evaluate the contributions of each variable in the

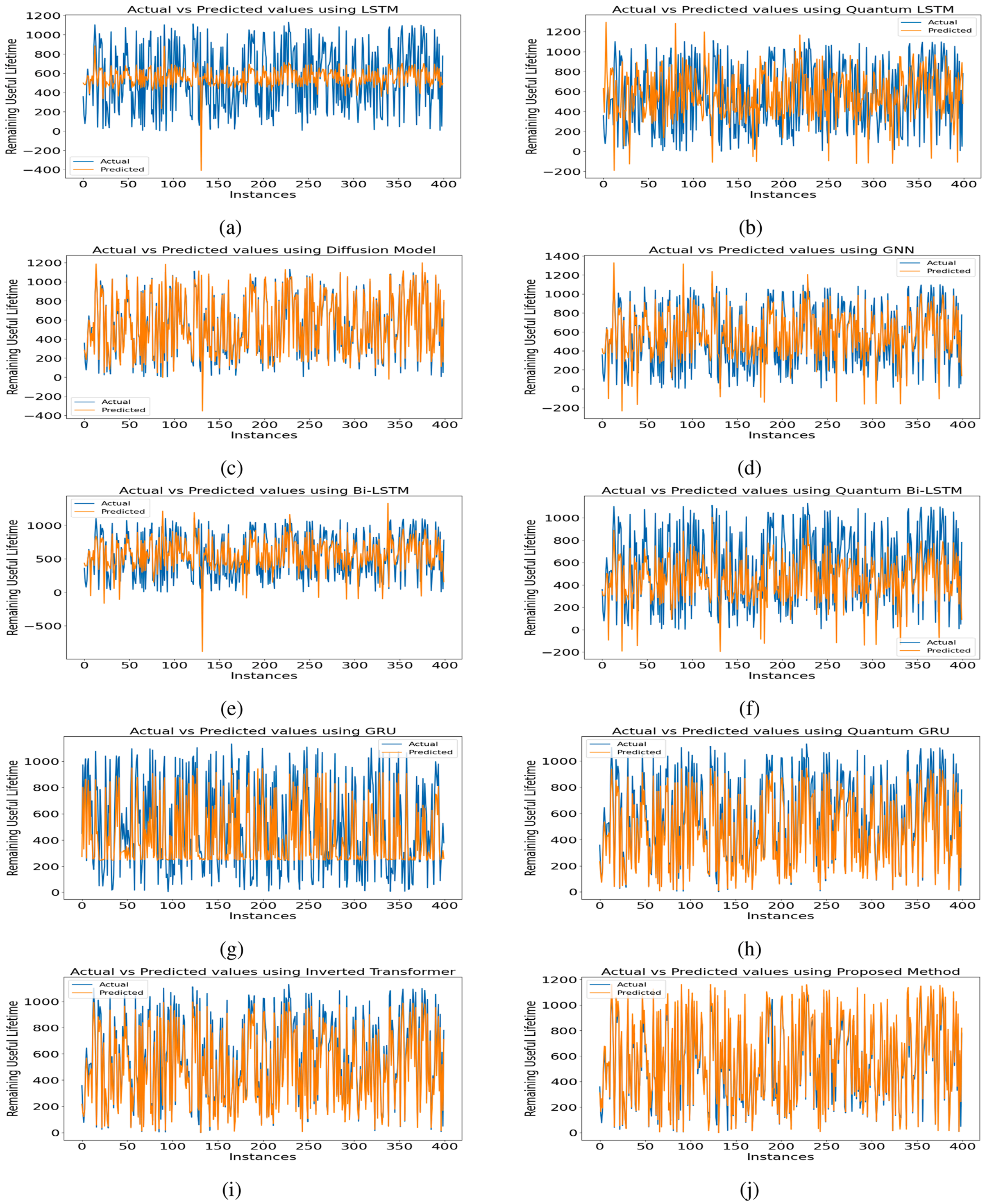


Fig. 4. Comparison of actual and predicted values across multiple models for the Predictive Maintenance dataset.

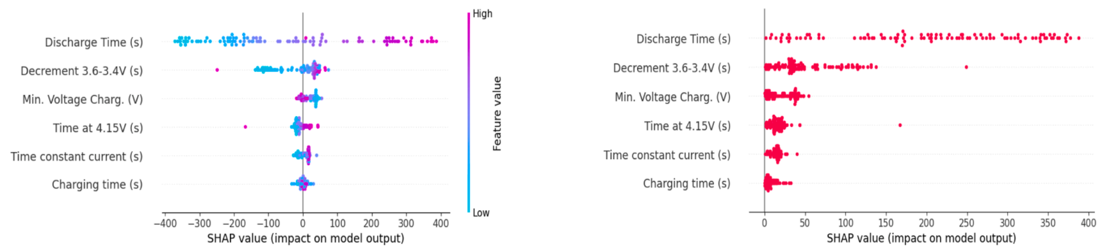
context of RUL prediction using the QiTransformer which could leading to a transparent understanding and feature significance.

The SHAP beeswarm plot (see Fig. 5a) enhances the understanding the magnitude and direction of the SHAP values with colors representing the feature values (ranging from low to high). In this plot *Discharge Time (s)* continues to show an impact where higher values (indicated in pink) tend to increase the output of the model whereas lower values (indicated in blue) tend to decrease it. For *Decrement 3.6-3.4V (s)* higher values display a mixed impact lean towards a negative influence. The other features including *Min. Voltage Charg. (V)*, *Time at 4.15V (s)*, *Time constant current (s)*, and *Charging time (s)* demonstrate variations in SHAP values based on their magnitude.

Fig. 5b represents the SHAP beeswarm absolute plot denotes the SHAP values of features and their impact. In this plot each dot

Table 5
Performance evaluation on Predictive Maintenance Dataset.

Model	MSE	RMSE	MAE	MAPE	Computation Time
LSTM [7]	913.36	30.22	64.22	79.02	50.36
Quantum LSTM [56]	801.89	28.31	58.05	68.12	48.27
Diffusion RUL [57]	748.16	27.35	55.26	62.44	69.49
Bi-LSTM [58]	618.54	24.87	47.96	52.78	52.42
Quantum Bi-LSTM [59]	565.50	23.78	51.68	30.11	50.12
GNN [60]	525.12	22.91	46.67	41.56	78.11
GRU [61]	426.78	20.65	53.18	33.31	45.21
Quantum GRU [62]	447.19	21.14	53.74	33.01	43.72
Inverted Transformer [50]	263.88	16.24	54.58	32.86	42.11
Quantum Inverted Transformer (Proposed)	**153.98**	**12.41**	**36.90**	**14.19**	**31.87**



(a) Beeswarm (b) Beeswarm absolute

Fig. 5. Explainability analysis showing (a) SHAP beeswarm plot and (b) absolute SHAP beeswarm plot.

corresponds to a prediction with the x axis showing the magnitude of the SHAP values. Features such as *Discharge Time (s)* exhibit an impact with SHAP values extending up to 400 indicating their influence. Other features like *Decrement 3.6-3.4V (s)*, *Min. Voltage Charg. (V)*, *Time constant current (s)*, *Time at 4.15V (s)*, and *Charging time (s)* shows considerable impacts which lower than that of *Discharge Time (s)*.

The SHAP decision plot (see Fig. 6a) shows the cumulative impact of the features of the model for a single prediction or a group of predictions. In this plot, the model output value centers around 600. *Discharge Time (s)* stands out with the highest impact significantly increasing the model output value. *Decrement 3.6-3.4V (s)* also contributes positively but to a lesser extent compared to Discharge Time. Features like *Time at 4.15V (s)* and *Time constant current (s)* demonstrate a decreasing impact on the model output. Meanwhile the features *Min. Voltage Charg. (V)* and *Charging time (s)* appear to have minimal or no impact on the final model output in this specific decision path. This plot is useful for understanding the contribution of each feature in a particular prediction instance by giving a clear path of how the final output is derived from the input features.

The SHAP summary bar plot (see Fig. 6b) displays the features mean absolute SHAP values. This plot denotes the average influence of each feature on the output of the model. The feature *Discharge Time (s)* has the highest impact with a mean(|SHAP value|) of 203.48 making it the most influential feature. *Decrement 3.6-3.4V (s)* follows with a significant but lower mean(|SHAP value|) of 51.88. Other features like *Time at 4.15V (s)*, *Min. Voltage Charg. (V)*, *Charging time (s)* and *Time constant current (s)* have progressively lower impacts.

The SHAP waterfall plot (see Fig. 7a) provides a detailed explanation of a single prediction by showing the individual contributions of each feature. The baseline value typically shown at the starting point of the plot is adjusted by the SHAP values of each feature to arrive at the final prediction. In this plot *Discharge Time (s)* significantly contributes to the prediction with a positive SHAP value of $+201.62$. On the other hand features like *Decrement 3.6-3.4V (s)*, *Time at 4.15V (s)*, *Time constant current (s)*, and *Min. Voltage Charg. (V)* contribute negatively by reducing the value of the prediction. *Charging time (s)* has a small positive contribution with a SHAP value of $+2.78$. Together these plots offer comprehensive insights into the decision making which revealing the most influential features and their specific impacts on the predictions of the model.

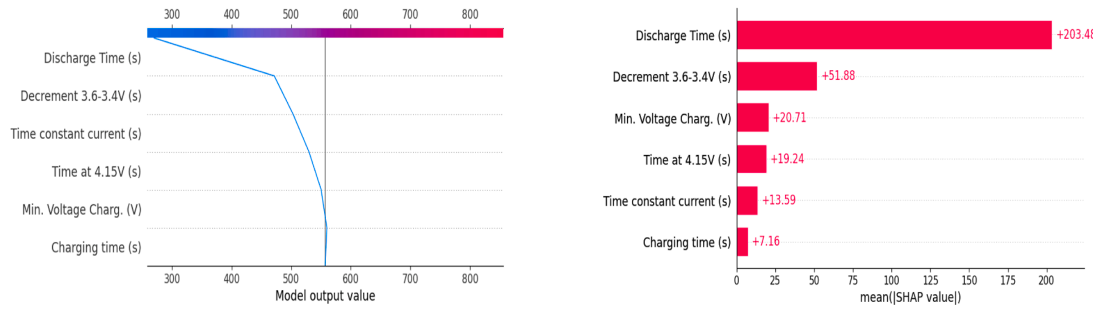
The SHAP bar cohort plot (see Fig. 7b) splits the data into two cohorts related to *Discharge Time (s)*. This plot highlights how the importance of feature varies between these cohorts. For the cohort where *Discharge Time (s)* $<$ 1608.75 then the feature maintains a high mean(|SHAP value|) of 208.49. In contrast for the cohort plot where *Discharge Time (s)* $>=$ 1608.75 the mean(|SHAP value|) slightly decreases to 201.05. The impact of other features like *Decrement 3.6-3.4V (s)* and *Time at 4.15V (s)* shows variation between the cohorts indicating how their importance shifts based on the discharge time split.

The SHAP heatmap plot (see Fig. 8a) displays the impact of each feature across instances. Features are represented in rows and instances are represented in columns. The color gradient from blue to red indicates the SHAP value showing how each feature influences the prediction for each instance. *Discharge Time (s)* stands out with variations in SHAP values indicating its dominant role in the model. This plot is useful for visualizing the consistency feature impacts across different instances providing insights into features. The SHAP violin plot (see Fig. 8b) which shows the distribution of SHAP values for each feature indicating their impact on output. *Discharge Time (s)* has the significant positive impact where higher values push the prediction higher. The width of the violin plot denotes the density of SHAP values. Positive contributions are visible for *Decrement 3.6-3.4V (s)*, *Min. Voltage Charg (V)*. The color gradient from red to blue indicates feature values from high to low.

The LIME plot (see Fig. 9) visualizes the features contribution to the estimation of the model. The predicted value is 39.70 with *Decrement 3.6-3.4 V (s)* being the significant feature negatively impacting the prediction. Features *Time constant current (s)*, *Discharge Time (s)*, and *Min. Voltage Charg (V)* contributes negatively to the prediction. The color indicates *Charging time (s)* as positive influencers (orange) and the rest as negative influencers (blue).

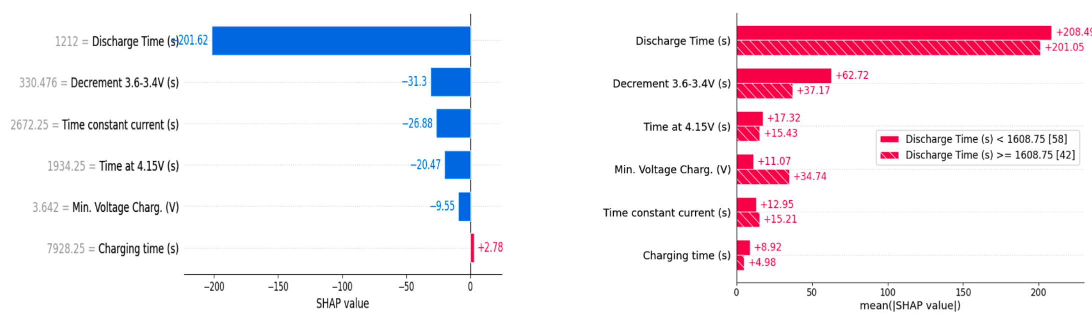
4.1.6. Counterfactual analysis on predictive maintenance dataset

Counterfactual analysis used for exploring "what-if" scenarios in predictive models allowing us to analyze how changes in input features can influence outcomes [63]. In our study we used a PdM dataset containing



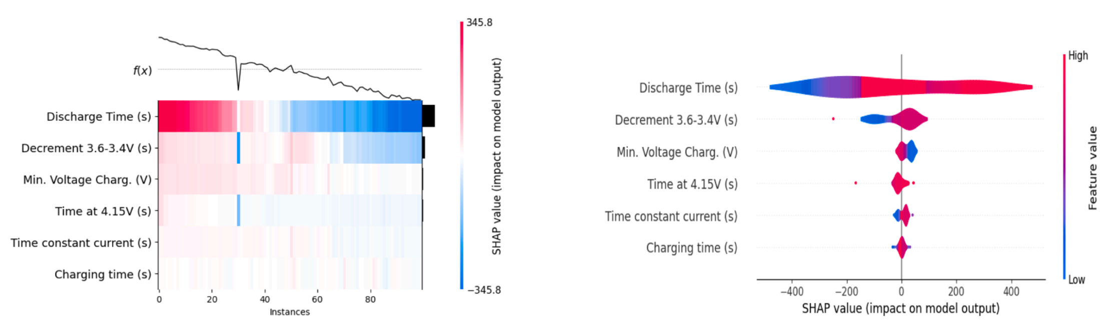
(a) (b)

Fig. 6. Explainability analysis showing (a) SHAP Decision Plot and (b) SHAP Bar Plot.



(a) (b)

Fig. 7. Explainability analysis showing (a) SHAP Waterfall Plot and (b) SHAP Cohort Plot.



(a) (b)

Fig. 8. Explainability analysis showing (a) SHAP Heatmap Plot and (b) SHAP Violin Plot.

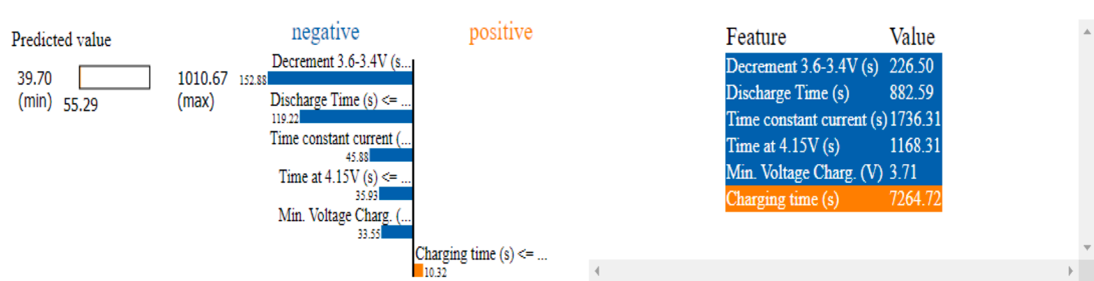


Fig. 9. LIME analysis on a selected instance from the Predictive Maintenance dataset.

parameters of battery performance including the *Time constant current (s)*, *Time at 4.15V (s)*, *Discharge Time (s)*, *Charging time (s)*, *Decrement 3.6-3.4V (s)*, and *Min. Voltage Charg (V)*. The target variable was *RUL* which varies by providing insights into the longevity of battery performance.

Starting with a query instance that had a predicted *RUL* of 266 s, we generated counterfactual examples to explore scenarios and how changes in parameter influence the *RUL* within the range of 0 to 1133 s. The first counterfactual showed that without specifying a change in parameters which is an alteration in the *Discharge Time (s)* to 751,147.8 s dramatically adjusts the scenario indicating a unrealistic or operational shift. The second scenario involved adjusting the *Time at 4.15V (S)* to 191,926.056 s while the *Min. Voltage Charg. (V)* was set to 3.413 volts resulting in an RUL prediction of 339.79 s. Another counterfactual

Table 6
Ablation Study on Predictive Maintenance Dataset.

Model	MSE	RMSE	MAE	MAPE	Computation Time
Without Causal Inference	1189.29	34.48	128.24	67.91	228.34
Without Embedding	573.10	23.93	110.87	51.27	80.65
Without Transformer Blocks	724.82	26.92	96.29	49.34	45.38
Without Layer Normalization	479.46	21.89	71.98	35.47	62.17
Without Dense Layer	313.92	17.71	51.27	33.21	57.31
Quantum Inverted Transformer (Proposed)	**153.98**	**12.41**	**36.90**	**14.19**	**31.87**

suggested a change in the *Charging time (s)* to 509,459.81 s which brought the *RUL* close to the original value at 254.77 s. Moreover, an increase in the *Time constant current (s)* to 571,153.57 s and the *Min. Voltage Charg. (V)* to 4.322 volts resulted in a longer *RUL* of 314.3 s. Similarly, altering the *Discharge Time (s)* to 73,314.9 s provided a new RUL of 307.75 s showing how modifications in a single parameter could affect battery life. Finally setting the *Min. Voltage Charg. (V)* to 3.884 volts yielded an RUL of 310.55 s.

These counterfactuals illustrate how specific changes to battery operational parameters can lead to diverse outcomes in terms of RUL predictions. The insights gained from analysis can guide targeted adjustments to enhance battery performance and longevity could supporting effective maintenance.

4.1.7. Ablation study on predictive maintenance dataset

An ablation study was carried out to assess the contribution of individual components within the proposed architecture. To this end, we evaluated several reduced variants of the model by selectively removing elements such as the causal inference module, embedding layer, transformer blocks, layer normalization, and dense layers. Table 6 summarises these configurations: *Without Causal Inference* indicates that no causative relationships were identified; *Without Embedding* refers to the exclusion of quantum embedding operations; *Without Transformer Blocks* denotes a model variant without transformer-based sequence processing; *Without Layer Normalization* corresponds to omitting the normalization stage; and *Without Dense Layer* signifies removal of the final dense layer. The findings show that the complete model consistently outperforms all ablated variants. The proposed architecture attained an MSE of 153.98, an RMSE of 12.41, and a MAE of 36.90, all of which are lower than the errors obtained when critical components were removed. Notably, excluding transformer blocks or layer normalization resulted in substantial performance degradation, emphasizing their role in capturing temporal structure within the data. Additionally, the full model achieved the shortest computation time of 31.87 s, demonstrating its efficiency and robustness for predictive maintenance applications. Table 6 presents the detailed outcomes of the ablation analysis.

4.2. Performance evaluation on C-MAPSS dataset

The C-MAPSS dataset is a collection of simulated turbofan engine degradation data from NASA, used for prognostics and health management (PHM) research to predict a turbofan engine's RUL. The dataset records 21 sensor measurements together with three operational settings for each engine, capturing their behaviour throughout different stages of operation. These time-series measurements form the basis for estimating the RUL of engines. The data is organised into training and test partitions: the training set contains complete engine lifecycles up to failure, whereas the test set includes sequences truncated before the point of failure. This structure enables the development and evaluation of models tasked with forecasting the remaining operational duration of an engine prior to required maintenance. As the dataset originates from the aviation domain, a sector characterised by stringent safety and regulatory requirements, the C-MAPSS data inherently reflects a controlled and compliant predictive maintenance environment. Its operational variables are shaped by industry safety norms and maintenance procedures, providing a realistic representation of regulated industrial conditions. A detailed overview of the dataset is provided in Table 7, while the description of individual variables is summarised in Table 8.

4.2.1. Data preprocessing on C-MAPSS dataset

The data preprocessing pipeline was carefully structured to enhance the accuracy and stability of the RUL prediction model for aircraft engines. We first verified the dataset for completeness, ensuring that no missing values were present, thereby preserving the reliability of the subsequent analysis. Following this, a series of feature engineering procedures were implemented to extract more informative representations from the raw sensor measurements and operational settings, enabling a more precise characterisation of engine degradation dynamics. All sensor variables were then normalised to mitigate disparities in scale and operating ranges, while operational settings were standardised to minimise the influence of varying flight conditions. In the final step, time-series data of each engine was partitioned into fixed-length sequences, a formulation that facilitates the capacity of the model to capture temporal dependencies essential for robust RUL estimation.

4.2.2. Causal inference on C-MAPSS dataset

We identified several key variables from LiNGAM method that have a direct causal influence on the RUL of aircraft engines, including critical such as *(HPC outlet Static pressure) (psia)'*, *(HPC outlet pressure) (psia)'*, *(Corrected fan speed) (rpm)*, *Bleed Enthalpy*, *(Physical core speed (rpm))*, *(Corrected core speed (rpm))*, and *(Physical fan speed) (rpm)'*. These variables are significant because they capture essential aspects of the operational state thereby providing the factors that affect the wear and tear over time of the engine. The causal directed acyclic graph on C-MAPSS dataset is showed in Fig. 10. By centering the model training on the identified causal variables, we were able to streamline the feature space and optimise computational efficiency. This selective approach reduced model complexity while preserving- and in some cases enhancing-predictive performance. Incorporating the full set of available variables would have introduced redundant or weakly informative features, potentially increasing noise and training time. In contrast, the focused feature set accelerated model convergence and improved interpretability, thereby facilitating practical deployment in real-world PdM environments. As a result, this optimisation strategy enabled more reliable and timely RUL predictions, supporting the reduction of unplanned downtime across engine fleets.

4.2.3. Actual versus predicted values on C-MAPSS dataset

The graphical representation of the actual versus predicted RUL values for various models is illustrated Fig. 11. Fig. 11a presents the performance of the LSTM. In Fig. 11b the quantum LSTM performance of the model is depicted to presents improvements in predictive accuracy compared to the LSTM. Similarly the Fig. 11c presents the GRU results while Fig. 11d illustrates the quantum GRU model which leverages quantum computing to enhance prediction precision.

Fig. 11e displays the predictions from the Bi-LSTM model used to capture bidirectional dependencies in the data. In Fig. 11f the quantum Bi-LSTM predictions are represented by demonstrating the gains achieved through quantum enhancements. Fig. 11g provides the results for the inverted transformer model by handling long range dependencies. Finally the Fig. 11h presents the quantum inverted transformer

Table 7
Summary of the C-MAPSS Dataset.

Feature	count	mean	std	min	25%	50%	75%	max
engine	20631.0	51.506568	29.22763	1.0000	26.0000	52.0000	77.0000	100.0000
cycle	20631.0	108.807862	68.88099	1.0000	52.0000	104.0000	156.0000	362.0000
setting_1	20631.0	−0.000009	0.002187	−0.0087	−0.0015	0.0000	0.0015	0.0087
setting_2	20631.0	0.000009	0.002930	−0.0100	−0.0000	0.0000	0.0000	0.0100
setting_3	20631.0	100.000000	0.000000	100.0000	100.0000	100.0000	100.0000	100.0000
(Fan inlet temperature) (°R)	20631.0	518.670007	0.000532	518.6700	518.6700	518.6700	518.6700	518.6700
(LPC outlet temperature) (°R)	20631.0	642.680934	5.000533	621.4100	642.5300	642.6400	643.8000	644.5300
(HPC outlet temperature) (°R)	20631.0	1590.523119	6.131150	1571.0400	1586.1900	1594.3800	1598.1600	1616.9100
(LPT outlet temperature) (°R)	20631.0	1408.933782	9.000605	1382.2500	1402.3600	1408.0400	1414.5550	1441.4900
(Fan inlet Pressure) (psia)	20631.0	14.620000	1.7764e-15	14.6200	14.6200	14.6200	14.6200	14.6200
(bypass-duct pressure) (psia)	20631.0	21.609803	1.3889e-05	21.6100	21.6100	21.6100	21.6100	21.6100
(HPC outlet pressure) (psia)	20631.0	553.377111	8.850932	549.8500	550.5600	552.8200	554.7100	556.6800
(Physical fan speed) (rpm)	20631.0	2388.096652	7.095842	2387.9000	2388.0400	2388.0400	2388.1400	2388.5600
(Physical core speed) (rpm)	20631.0	9065.242941	20.88288	9021.7300	9053.0000	9060.6600	9069.4200	9244.5900
(Engine pressure ratio (P50/P2))	20631.0	1.300000	0.000000	1.3000	1.3000	1.3000	1.3000	1.3000
(HPC outlet Static pressure) (psia)	20631.0	47.541168	2.670874	46.8500	47.3500	47.5100	47.7000	48.5300
(Ratio of fuel flow to Ps30) (pps/psia)	20631.0	521.413470	7.375534	518.6900	520.9600	521.4800	521.9500	523.8300
(Corrected fan speed) (rpm)	20631.0	2388.096152	1.791829	2378.8800	2387.8800	2388.0400	2388.1400	2388.7200
(Corrected core speed) (rpm)	20631.0	8143.175272	1.991214	8099.3100	8141.7800	8143.8000	8144.3700	8389.7200
(Bypass Ratio)	20631.0	8.442146	0.037505	8.3249	8.4149	8.4389	8.4656	8.5484
(Burner fuel-air ratio)	20631.0	0.031000	1.387821e-17	0.0310	0.0310	0.0310	0.0310	0.0310
(Bleed Enthalpy)	20631.0	393.210654	1.548763	392.0000	392.0300	394.0000	394.0200	394.0500
(Required fan speed)	20631.0	2388.000000	0.000000	2388.0000	2388.0000	2388.0000	2388.0000	2388.0000
(Required fan conversion speed)	20631.0	100.000000	0.000000	100.0000	100.0000	100.0000	100.0000	100.0000
(High-pressure turbines Cool air flow)	20631.0	38.816217	1.050865	37.2000	37.7700	38.6700	39.5200	39.8800
(Low-pressure turbines Cool air flow)	20631.0	23.289705	1.082509	22.8942	23.1218	23.2218	23.3668	23.6184

Table 8
Summary of variables in the C-MAPSS dataset.

Variable Name	Description	Variable Type	Environment Characteristic
engine	Identifier for each engine unit	Categorical	Regulated
Cycle	Time step corresponding to each engine's operational cycle	Sequential	Regulated
setting_1	Operational setting related to flight condition 1	Continuous	Regulated
setting_2	Operational setting related to flight condition 2	Continuous	Regulated
setting_3	Operational setting related to flight condition 3	Continuous	Regulated
(Fan inlet temperature) (°R)	Total temperature at fan inlet	Continuous	Regulated
(LPC outlet temperature) (°R)	Total temperature at low-pressure compressor outlet	Continuous	Regulated
(HPC outlet temperature) (°R)	Total temperature at high-pressure compressor outlet	Continuous	Regulated
(LPT outlet temperature) (°R)	Total temperature at low-pressure turbine outlet	Continuous	Regulated
(Fan inlet Pressure) (psia)	Pressure at fan inlet	Continuous	Regulated
(bypass-duct pressure) (psia)	Pressure in bypass-duct	Continuous	Regulated
(HPC outlet pressure) (psia)	Pressure at high-pressure compressor outlet	Continuous	Regulated
(Physical fan speed) (rpm)	Measured fan rotational speed	Continuous	Regulated
(Physical core speed) (rpm)	Measured core rotational speed	Continuous	Regulated
(Engine pressure ratio (P50/P2))	Ratio of exhaust to inlet pressure	Continuous	Regulated
(HPC outlet Static pressure) (psia)	Static pressure at high-pressure compressor outlet	Continuous	Regulated
(Ratio of fuel flow to Ps30) (pps/psia)	Fuel flow ratio relative to compressor discharge pressure	Continuous	Regulated
(Corrected fan speed) (rpm)	Corrected fan speed for environmental conditions	Continuous	Regulated
(Corrected core speed) (rpm)	Corrected core speed for environmental conditions	Continuous	Regulated
(Bypass Ratio)	Ratio of bypass air mass flow to core flow	Continuous	Regulated
(Burner fuel-air ratio)	Ratio of fuel to air mixture in burner	Continuous	Regulated
(Bleed Enthalpy)	Enthalpy of bled air from the engine	Continuous	Regulated
(Required fan speed)	Target rotational speed of the fan section	Continuous	Regulated
(Required fan conversion speed)	Normalized fan speed used for control and monitoring	Continuous	Regulated
(High-pressure turbines Cool air flow)	high pressure turbine to maintain blade temperature limits	Continuous	Regulated
(Low-pressure turbines Cool air flow)	low pressure turbine to maintain blade temperature limits	Continuous	Regulated

(proposed method) predictions of the model underscoring its superior performance in aligning with the actual RUL values thereby confirming its efficacy and robustness in predicting the RUL of the devices.

4.2.4. Performance evaluation on C-MAPSS dataset

In our comparative evaluation of models for forecasting the RUL of aircraft engines using the NASA C-MAPSS dataset, the findings demonstrate a distinct performance advantage of quantum-enhanced architectures over their classical counterparts (see Table 9). All models shown in Table 9 were re-implemented under the same preprocessing pipeline, feature configuration, sequence construction, and train-test split that were used for the QiTransformer to ensure a controlled and fair comparison. The standard LSTM model [7] with an MSE of 483.56 and RMSE of 21.98 demonstrated accuracy but the quantum LSTM [56] showed improved performance by reducing these errors and yielding an MSE of 408.63. The Bi-LSTM [58] and GRU [61] models further improved accuracy with their quantum versions then the quantum Bi-LSTM [59] and quantum GRU [62] showing greater reductions in error metrics. The GNN model [60] exhibited performance by capturing relational dependencies within the data by achieving an MSE of 256.08 an RMSE of 16.00 and a MAPE of 2.54. DiffusionRUL [57] which models degradation trajectories through diffusion based generative processes achieved competitive results with an MSE of 248.11 RMSE of 15.75 and a MAPE of 2.41. The classical inverted transformer [50] achieved an MSE of 197.29

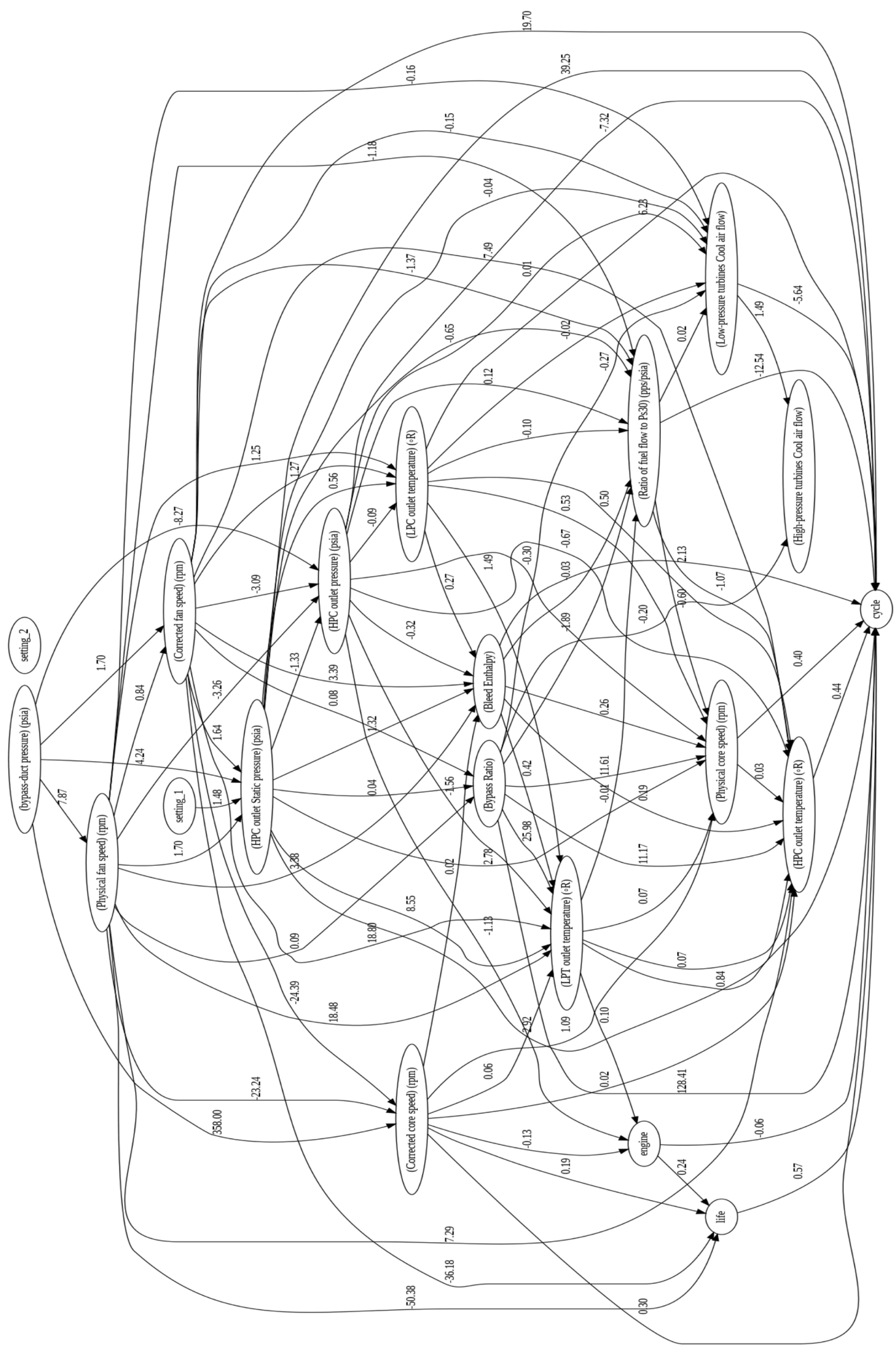


Fig. 10. Causal Directed Acyclic Graph (DAG) illustrating the causal relationships governing the C-MAPSS Dataset.

and an RMSE of 14.04. The proposed quantum inverted transformer outperformed all other models and achieved the lowest MSE of 96.42 RMSE of 9.81 and MAPE of 0.78 while maintaining efficient computation time at 36.72 s.

4.2.5. Explainability analysis on C-MAPSS dataset

The SHAP beeswarm plot (see Fig. 12a) visualizes the impact of various features on the predictions for the RUL of aircraft engines. Each dot represents an individual prediction instance positioned along the x axis based on its SHAP value which indicates the feature contribution to the predicted RUL. The dots are color coded from blue (low feature values) to pink (high feature values) represents the influence of the feature magnitude. Features like *(Corrected core speed)(rpm)* and *(Physical core speed)(rpm)* display a spread of SHAP values suggesting they have a impact on the predictions of the model. In contrast the features such as *(HPC outlet pressure) (psia)* show a spread indicating a consistent but smaller effect. This plot underscores the role of core and fan speed parameters in influencing the predicted RUL with high values increasing the predicted life span.

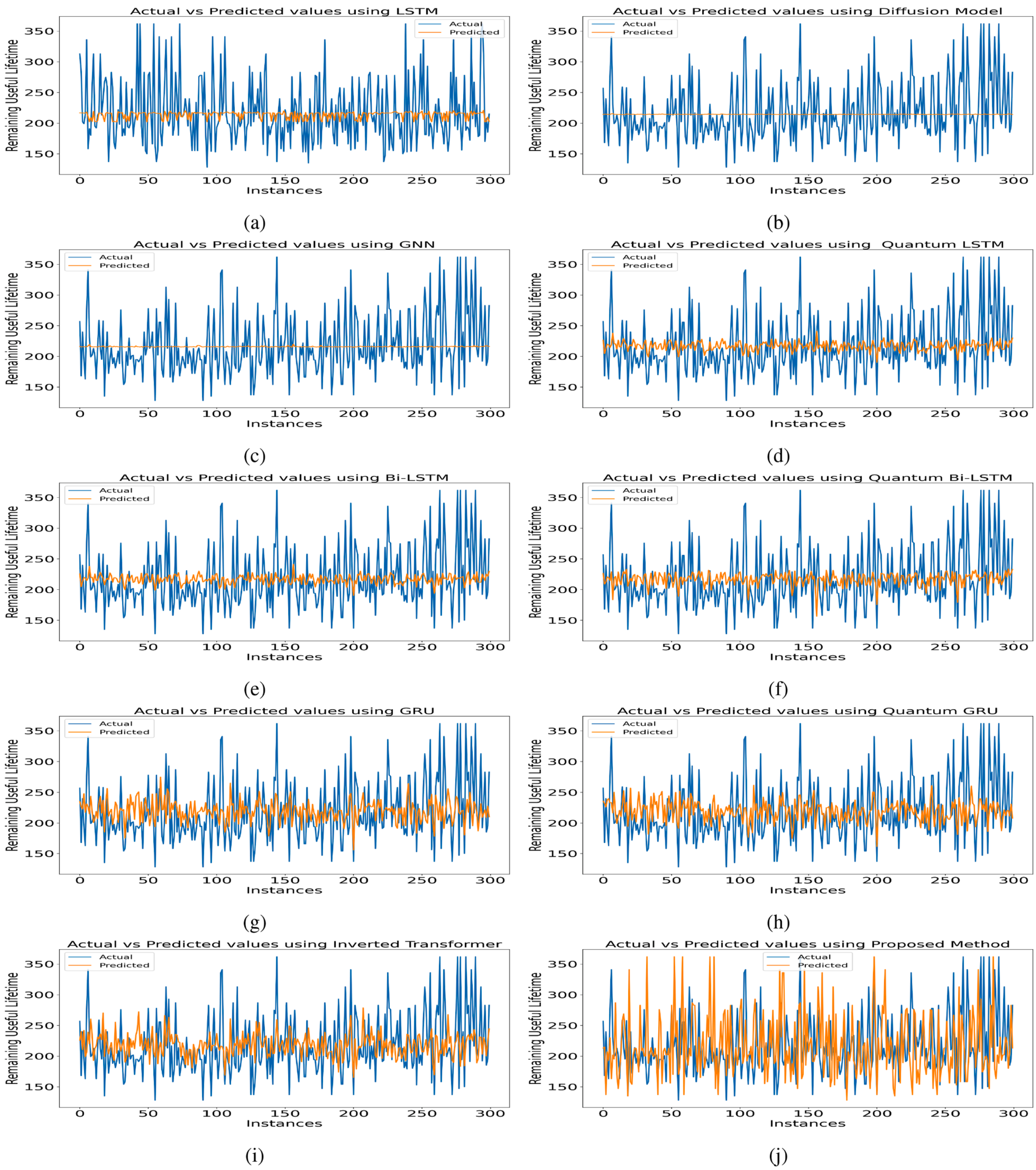


Fig. 11. Comparison of actual and predicted values across multiple models for the C-MAPSS Dataset.

Table 9
Evaluation of various metrics using various models on C-MAPSS Dataset.

Model	MSE	RMSE	MAE	MAPE	Computation Time
LSTM [7]	483.56	21.98	59.21	6.11	42.15
Quantum LSTM [56]	408.63	20.21	51.71	5.46	48.22
Bi-LSTM [58]	269.84	16.42	39.18	2.59	71.41
GNN [60]	256.08	16.00	36.66	2.54	78.43
Diffusion RUL [57]	248.11	15.75	35.26	2.41	61.34
Quantum Bi-LSTM [59]	212.49	14.57	32.89	1.67	68.88
GRU [61]	304.58	17.45	43.54	3.44	51.96
Quantum GRU [62]	272.83	16.51	37.97	2.07	40.21
Inverted Transformer [50]	197.29	14.04	31.55	1.41	38.11
Quantum Inverted Transformer (Proposed)	**96.42**	**9.81**	**15.21**	**0.78**	**36.72**

Table 10
Ablation Study on C-MAPSS Dataset.

Model	MSE	RMSE	MAE	MAPE	Computation Time
Without Causal Inference	746.62	27.32	64.89	15.27	296.37
Without Embedding	452.18	21.26	41.27	13.79	145.85
Without Transformer Blocks	520.24	22.80	38.62	12.01	120.42
Without Layer Normalization	225.14	15.00	28.96	16.87	142.85
Without Dense Layer	196.53	14.01	22.54	12.19	61.42
Quantum Inverted Transformer (Proposed)	**96.42**	**9.81**	**15.21**	**0.78**	**36.72**

Table 11
Significance of Friedman and Nemenyi test.

Model	Average Rank	Difference	Significance (Y / N)
LSTM	9.1	8.1	Y
Quantum LSTM	8.2	7.2	Y
Bi-LSTM	6.7	5.7	Y
Quantum Bi-LSTM	4.4	3.4	Y
GRU	5.8	4.8	Y
Quantum GRU	4.8	3.8	Y
GNN	5.9	4.9	Y
Diffusion RUL	6.5	5.5	Y
Inverted Transformer	2.6	1.6	N
Quantum Inverted Transformer	1.0	NA	NA

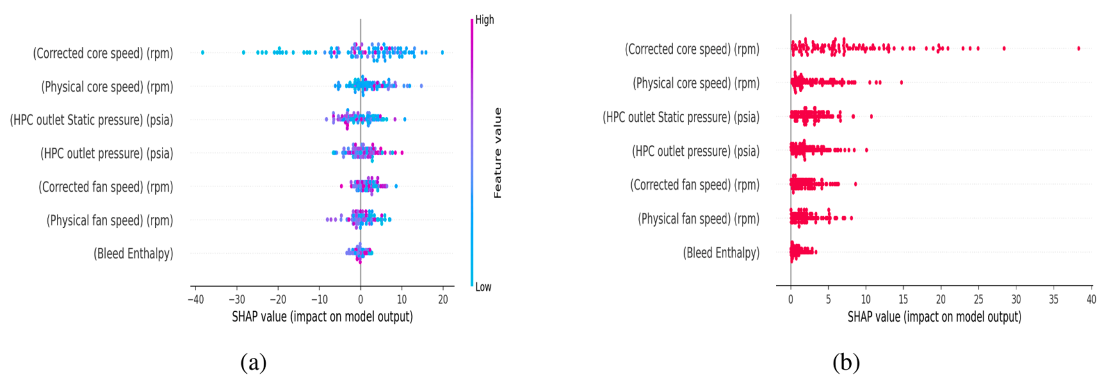


Fig. 12. Explainability analysis showing (a) SHAP beeswarm plot and (b) absolute SHAP beeswarm plot.

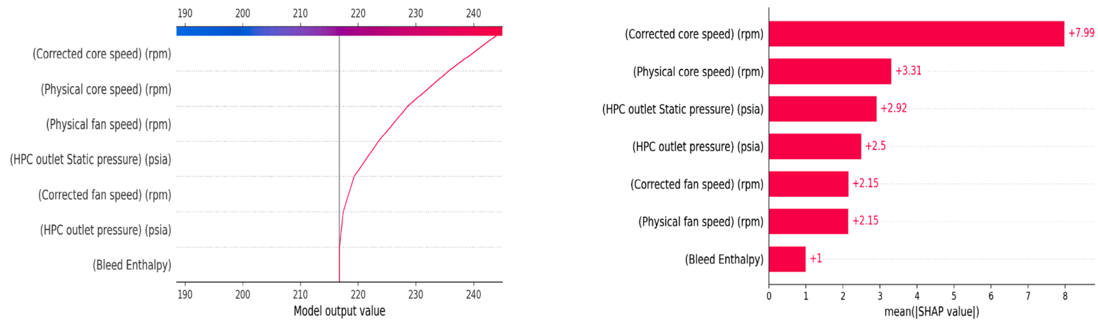
The SHAP beeswarm plot (see Fig. 12b) focuses on the absolute values of SHAP scores which helps in assessing the significance of each feature in the predictions of the model. The features with higher absolute SHAP values like *(Corrected core speed) (rpm)* and *(Physical core speed) (rpm)* are the influential as indicated by their wide spread across the x axis. This demonstrates that these features consistently play a significant role in the predictions of the model. Other features such as *(HPC outlet Static pressure) (psia)* and *(Corrected fan speed) (rpm)* while showing a smaller range of impacts still contribute uniformly to the prediction process.

The SHAP decision plot (see Fig. 13a) illustrates how each feature cumulatively impacts the prediction of RUL for aircraft engines. On the x axis the model output values range from 190 to 240 representing the predicted RUL in cycles. The y axis lists the features ordered by their influence. The color gradient of the plot from blue (low values) to pink (high values) shows how the feature values correlate with the RUL prediction. Corrected Core Speed (rpm) has the most substantial impact and significantly increasing the RUL as its value rises which starting from a baseline prediction around 190. *(Physical Core Speed) (rpm)* and *(HPC Outlet Static Pressure) (psia)* contribute positively pushing the RUL higher. Features like *(HPC Outlet Pressure) (psia)* and *(Corrected Fan Speed) (rpm)* offer moderate adjustments to refining the estimation of the model. *Physical Fan Speed (rpm)* and *Bleed Enthalpy* have smaller effects but still influence the final prediction. This plot visually breaks down how each feature shapes the *RUL* estimate thereby providing insights into the decision making for operational scenario of each engine.

The bar plot (see Fig. 13b) displays the mean absolute SHAP values for features highlighting their importance in the model. The feature with the highest influence is the corrected core speed (rpm) with a SHAP value of 7.99 indicating it has the significant impact on the predictions of the model. Following this the *(physical core speed) (rpm)* holds a SHAP value of 3.31 making it the second most influential feature. Other features like *(HPC outlet Static pressure) (psia)* and *(corrected fan speed) (rpm)* contribute significantly and albeit to a lesser extent. This plot effectively ranks the features by their average impact with the most influential ones listed at the top.

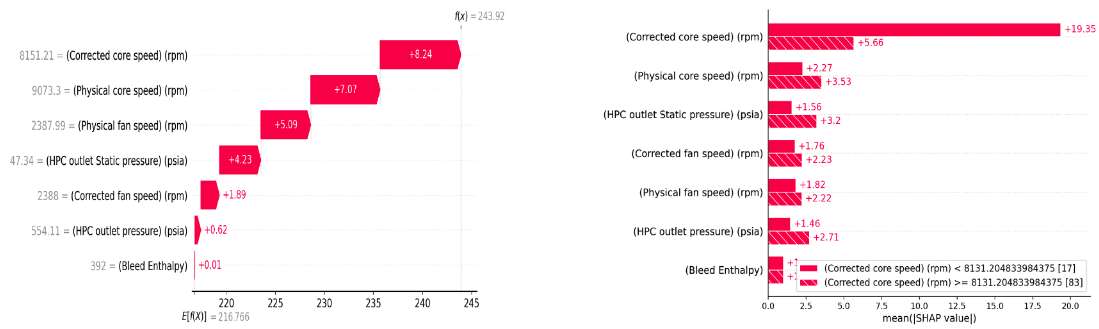
The waterfall plot (see Fig. 14a) that provides an explanation for a single prediction illustrating how individual features contribute to the final predicted value. The expected value is 216.77 and the final prediction is 243.92. The *(corrected core speed) (rpm)* at 8151.21 adds to the prediction with an increase of 8.24. Similarly the *(physical core speed) (rpm)* at 9073.3 and the *(physical fan speed) (rpm)* at 2387.99 add 7.07 and 5.09 respectively. Other features like *(HPC outlet Static pressure) (psia)* and *(corrected fan speed) (rpm)* contribute additional increases and albeit smaller. This plot provides a detailed about how each feature drives the prediction from the base value to the final output offering a clear explanation of the decision making.

The cohort plot (see Fig. 14b) compares the SHAP values for the *(corrected core speed) (rpm)* feature under different conditions or cohorts. Specifically it divides the *(corrected core speed) (rpm)* into two cohorts based on whether the value is below or above 8131.20. For



(a) (b)

Fig. 13. Explainability analysis showing (a) SHAP Decision Plot and (b) SHAP Bar Plot.



(a) (b)

Fig. 14. Explainability analysis showing (a) SHAP Waterfall Plot and (b) SHAP Cohort Plot.

values below this threshold the SHAP value is 5.66 while for values above it skyrockets to 19.35 indicating a higher influence in the second cohort. This comparison extends to features like *(physical core speed) (rpm)* and *(HPC outlet Static pressure) (psia)* which varying SHAP values across the cohorts revealing how feature importance changes with various value ranges.

The SHAP heatmap (see Fig. 15a) provides a visual representation of the SHAP values for each feature across different instances using a color scale to represent the impact of each feature. In this heatmap red indicates a high positive impact and blue indicates a high negative impact. The features in the heatmap are *(Corrected core speed) (rpm)*, *(Physical core speed) (rpm)*, *(HPC outlet Static pressure) (psia)*, *(HPC outlet pressure) (psia)*, *(Corrected fan speed) (rpm)*, *(Physical fan speed) (rpm)*, and *Bleed Enthalpy*. This visualization allows how each feature influences the prediction for instances with the color intensity providing a clear direction of impact of each feature.

The violin plot (see Fig. 15b) illustrates the distribution of SHAP values for each feature insights into the density and range of these values. The features displayed include *(Corrected core speed) (rpm)*, *(Physical core speed) (rpm)*, *(HPC outlet Static pressure) (psia)*, *(HPC outlet pressure) (psia)*, *(Corrected fan speed) (rpm)*, *(Physical fan speed) (rpm)*, and *Bleed Enthalpy*. The width of each violin plot at a given SHAP value represents the density of that value showing where SHAP values are concentrated. Additionally the color scale within the plot indicates high values of features represented in red and low values in blue. This plot helps in analyzing the impact of each feature on the output by showing how the SHAP values for each feature are distributed and highlighting their influence on the predictions.

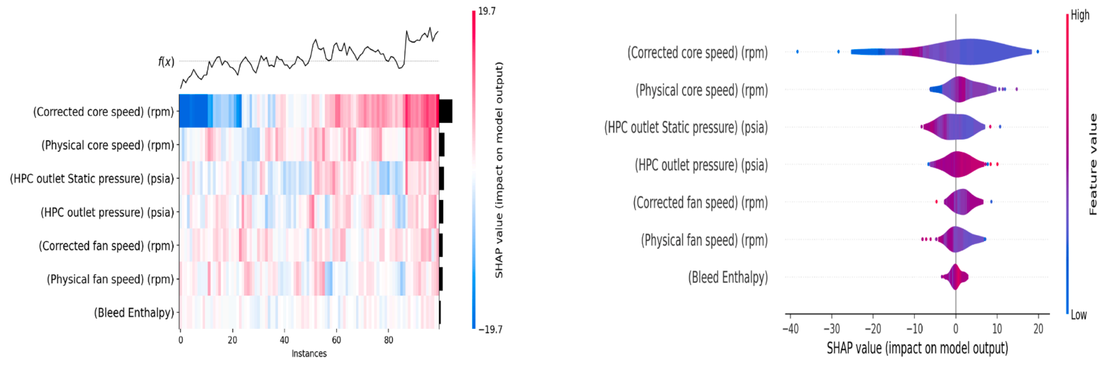
The Lime plot (see Fig. 16) provides an overview of the predicted value and the contribution of each feature. The model predicts a value of 243.87. The plot distinguishes between features that have a positive impact (highlighted in orange) and those with a negative impact (highlighted in blue). Among the features the *Corrected core speed* (8140.58) has the significant positive contribution of 4.85 followed by *Bleed Enthalpy* ($<=$ 392.00) with a contribution of 1.25. On the negative side *(HPC outlet pressure) (psia)* (552.80) reduces the prediction by 1.67 *(Physical fan speed) (rpm)* (2388.05) by 1.31 *(HPC outlet Static pressure) (psia)* (47.35) by 1.22 and *(Physical core speed) (rpm)* (9060.69) by 0.21. This breakdown helps in understanding which features are driving the predictions and their respective magnitudes.

4.2.6. Counterfactual analysis on C-MAPSS dataset

We examined a C-MAPSS dataset with features *(HPC outlet Static pressure) (psia)*, *(HPC outlet pressure) (psia)*, *(Corrected fan speed) (rpm)* and other measurements. Starting with an instance that had a predicted RUL of 240 h and we generated counterfactuals aiming to adjust this RUL within the desired range of 185 to 250 h. One counterfactual revealed that increasing the *(HPC outlet Static pressure) (psia)* from 47.38 to 48.48 would reduce the predicted RUL to 231.83 h. Another counterfactual indicated that the RUL could vary to 214.11 h without altering any parameters. These findings suggest that adjustments in specific features can impact the predicted RUL thereby offering valuable insights for proactive maintenance.

4.2.7. Ablation results on C-MAPSS dataset

We conducted a comprehensive ablation study to evaluate the contribution of each component within the proposed QiTransformer architecture for RUL prediction of aircraft engines. The removal of the causal inference module led to a pronounced deterioration in performance, with the MSE increasing to 746.62 and the RMSE to 27.32, underscoring the importance of identifying causative features rather than features rather than relying solely on associative patterns. Excluding the embedding layer similarly degraded the model, yielding an MSE of 452.18 and an RMSE of 21.26, indicating its essential role in enabling



(a) (b)

Fig. 15. Explainability analysis showing (a) SHAP Heatmap Plot and (b) SHAP Violin Plot.

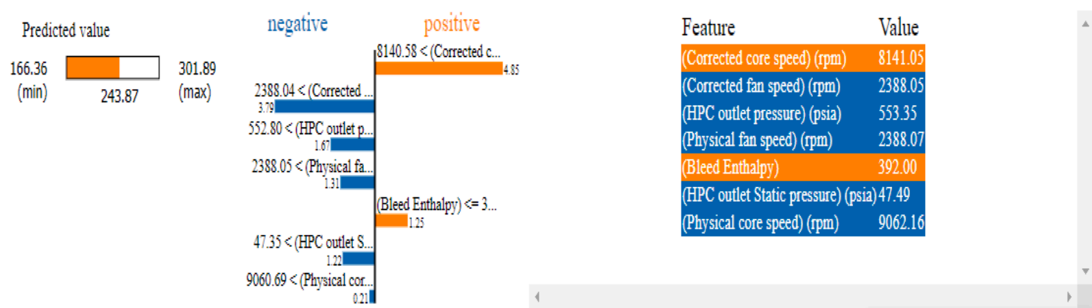


Fig. 16. LIME analysis on a selected instance from the C-MAPSS Dataset.

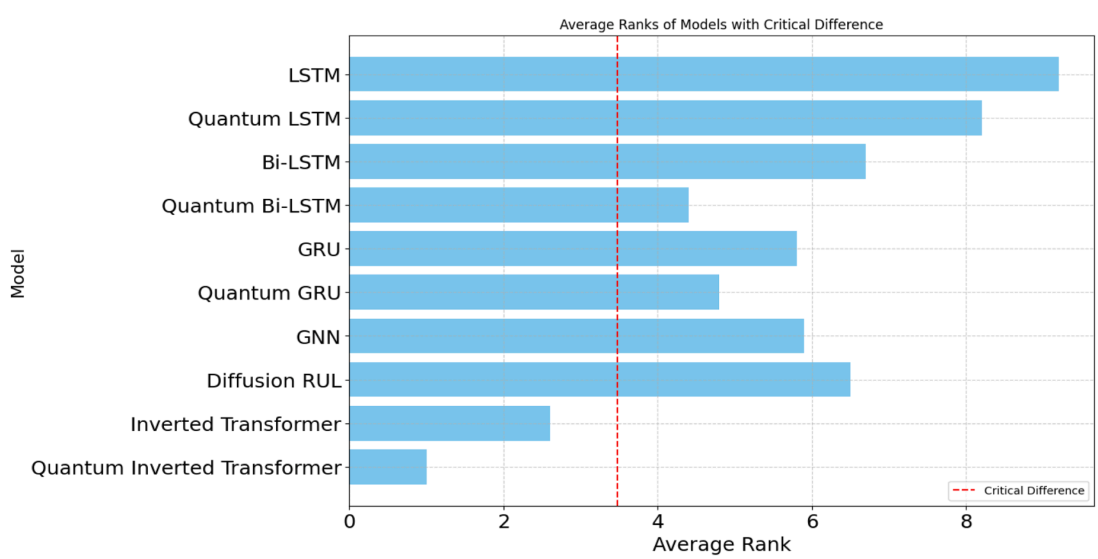


Fig. 17. Average model ranks from the Friedman test with the Nemenyi Critical Difference (CD).

effective representation of sensor and operational data. Eliminating the transformer block resulted in an MSE of 520.24 and an RMSE of 22.80, demonstrating that these blocks are critical for modelling temporal dependencies inherent in engine degradation. The absence of layer normalization, which stabilises the training process, further increased the MSE to 225.14 and the RMSE to 15.00, highlighting its contribution to consistent model convergence. We also assessed a configuration without the dense layer responsible for integrating the learned feature representations. This variant produced an MSE of 196.53 and an RMSE of 14.01, confirming that the dense layer remains important for refining the final predictive output. Across all configurations, the complete QiTransformer achieved the best performance, delivering the lowest MSE (96.42), RMSE (9.81), and MAPE (0.78), along with a computation time of 36.72 s. These results collectively emphasise the necessity of each component-causal inference, embedding mechanisms, transformer blocks, layer normalization, and dense layers-in achieving high predictive accuracy and computational efficiency. The detailed outcomes of the ablation study are reported in Table 10.

4.3. Statistical analysis

To evaluate the relative performance of the models reported in the two result tables, we conducted a set of non-parametric statistical tests to determine whether the observed differences in their performance metrics are statistically meaningful. Because the same group of models is assessed across multiple metrics, the Friedman test is appropriate for detecting overall differences in performance. When the Friedman test indicates significance, the Nemenyi post hoc procedure is then used to identify the specific model pairs that differ. The analysis followed three steps: (1) the performance values from Tables 5 and 9 were collated and organised for statistical comparison; (2) the Friedman test was applied to assess whether the models exhibit statistically distinguishable performance across the metrics; and (3) when significance was detected, the Nemenyi test was used to pinpoint the pairs of models responsible for these differences.

The Friedman test produced a test statistic of 81.99 with a corresponding p-value of 0.12×10^{-10}. As this value is well below the conventional significance threshold of 0.05, we conclude that the models do not perform equivalently across the evaluated metrics. Consequently, the Nemenyi post hoc test was carried out. The average ranks for all models, listed in Table 11, were used to compute the critical difference (CD), which was found to be 3.478. This CD value specifies the minimum separation between any two average ranks for their performance difference to be regarded as statistically significant (see Table 11).

The results of the Friedman test indicate that the Quantum Inverted Transformer consistently ranks highest among all evaluated models, suggesting that it offers the strongest overall performance for predictive maintenance and RUL estimation. The subsequent Nemenyi post hoc analysis further supports this finding, showing that both the Quantum Inverted Transformer and the Inverted Transformer achieve significantly better performance than the remaining models. These outcomes highlight the value of integrating quantum-inspired enhancements into deep learning architectures for complex prognostics tasks. The average model ranks, along with the corresponding critical difference threshold, are illustrated in Fig. 17.

5. Conclusion and future work

This study presented the QiTransformer, a quantum-inspired inverted transformer architecture that combines causal inference and explainable artificial intelligence for RUL prediction in predictive maintenance settings. By using LiNGAM-based causal pruning, the framework selects a compact set of truly influential variables, which reduces input dimensionality and improves the efficiency of the quantum embedding stage. Quantum attention layers are then employed to capture temporal dependencies in the selected features, while the integrated SHAP-, LIME-, and counterfactual-based analysis provides rich explanations of how causal variables shape the predicted RUL.

The proposed approach was evaluated on two benchmark datasets: a battery degradation dataset and the NASA C-MAPSS turbofan engine dataset. Across both cases, the QiTransformer achieved higher prediction accuracy and lower computational cost than several strong baseline models, indicating that the combination of causal selection and quantum-inspired attention can provide a more compact yet expressive representation for RUL estimation. The explainability analysis highlighted *Discharge Time (s)* as the most influential factor for the battery data and *Corrected Core Speed (rpm)]* as the dominant driver for the C-MAPSS trajectories, demonstrating that the framework can not only forecast RUL but also expose the key physical drivers behind the predictions. These properties make the model suitable for safety- and reliability-critical maintenance decisions, where both performance and interpretability are essential.

This study opens several promising avenues for future research. First, the QiTransformer will be systematically adapted and validated across a broader range of industrial assets to rigorously assess its robustness under heterogeneous operating regimes and fault patterns. Second, the causal inference layer can be further refined by exploring alternative causal discovery algorithms and richer structural priors, with the objective of enhancing stability in noisy and nonstationary environments. Third, future work will concentrate on improving the deployability of quantum deep learning models in real-world settings, including the design of hardware-efficient quantum circuits, the use of approximate quantum simulators, and the development of hybrid quantum-classical schemes that reduce computational overhead without sacrificing predictive performance. In parallel, online and incremental learning strategies will be investigated so that the QiTransformer can adapt to evolving operating conditions and degradation behaviours without full retraining, which is essential for long-lived industrial assets.

Declarations

- In this study, we utilized two distinct datasets that are publicly accessible.
- Code availability (software application or custom code) Custom code developed.
- Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.
- Conflict of interest Authors declare that they have no conflict of interest.

CRediT authorship contribution statement

Manoranjan Gandhudi: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Alphonse P.J.A.:** Writing – review & editing, Methodology; **Sharan Srinivas:** Writing – review & editing, Methodology; **Gangadharan G.R.:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Data availability

Datasets used in this article are publicly available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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